Natural Computing

Informatica 2.2 – Computational Intelligence

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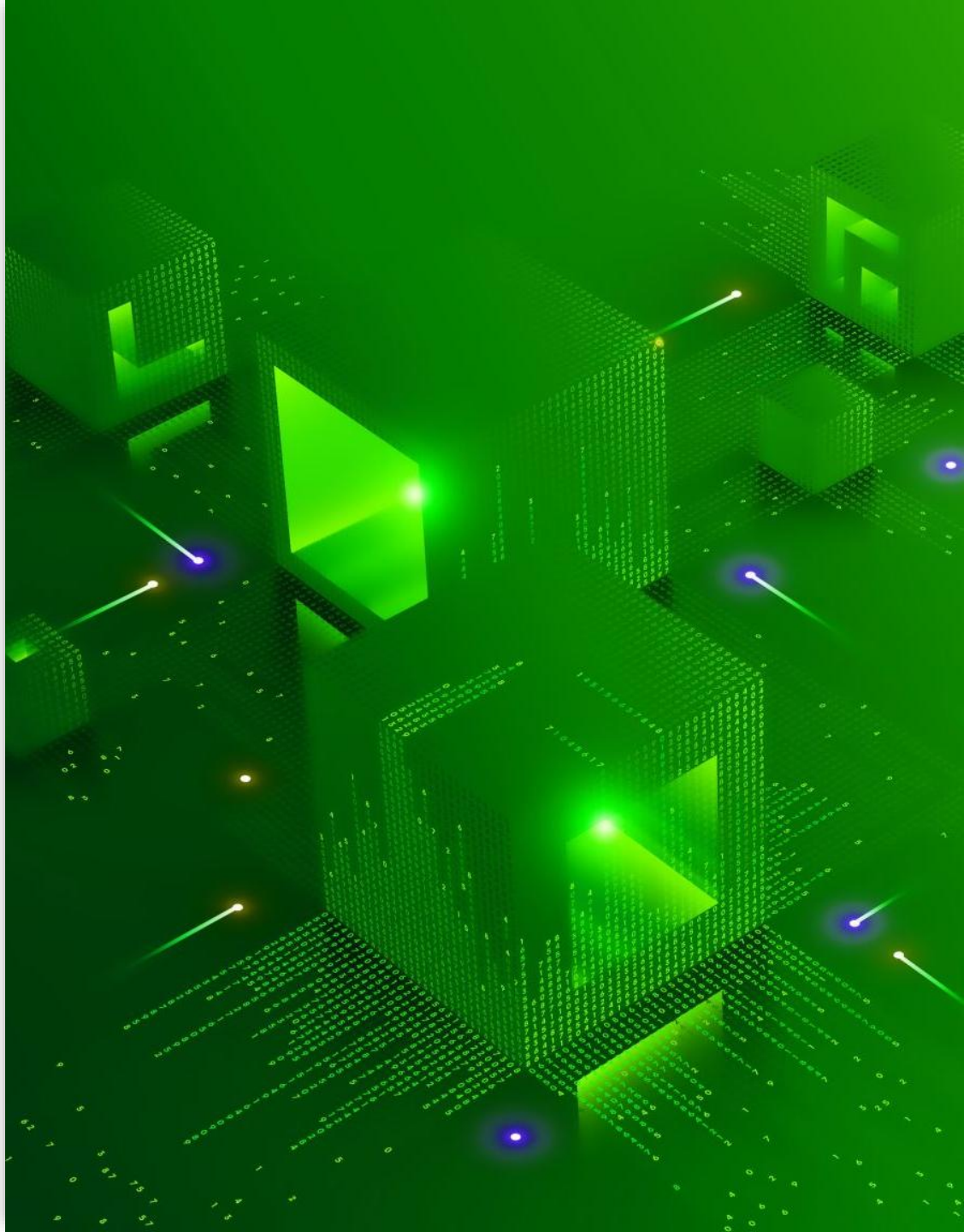
Natural Computing (NC)?

NC (aka Natural Computation) encompasses three classes of methods:

1) Methods taking inspiration from Nature (e.g., evolutionary algorithms, swarm intelligence, artificial neural networks)
2) Methods using computers to synthesize natural phenomena
3) **Methods that employ natural materials to perform computation**



Outline

- P vs NP
- NP-complete problems
- Turing Machines
- Deterministic vs Non-Deterministic Turing Machines
- DNA computing
  - Adleman molecular computer
  - Wang Tiles
  - Programmable DNA machines
- DNA Origami and DNA bricks

P

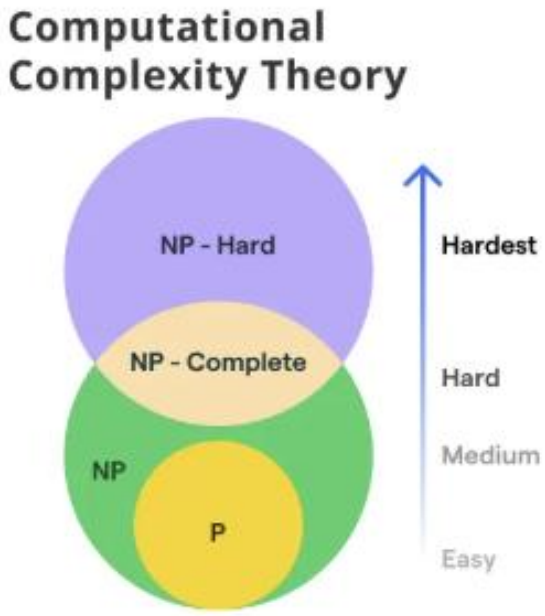
- **Computational complexity** is the amount of **resources** (time, or space) **needed to run an algorithm**
  - One important class is called «P»

- Informally, P contains all (decision) problems that can be **solved efficiently**
  - By efficiently we mean that it is solved by using an amount of computation time that is **polynomial with respect to the input**
  - By **decision problem** we mean all problems that can be posed as a **Y/N question**

- A lot of common problems belong to **P**

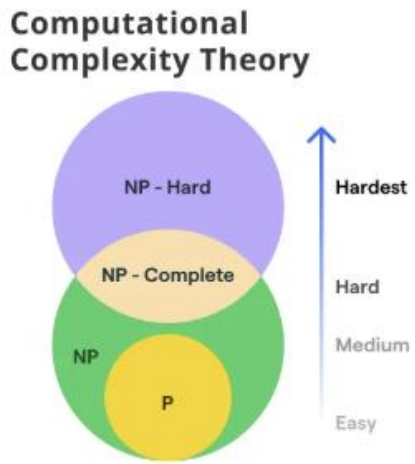
NP

- There are decision problems that **do not have an efficient algorithm, yet**
- Many of them belong to the **NP class of complexity**
- It is not know, at the moment, whether $P \neq NP$
  - Most computer scientists **think they are not equal**, but there is **no proof**
  - It is one of the **Millenium Prize Problems**, if you find a demonstration you win 1 million USD



NP-complete problems

- **Decision problems** belonging to the **NP class of complexity**
- Some of the **hardest in computer science**
- Intractable: **no algorithms are known**… at least on common computers (i.e., **deterministic Turing machines**)
- **Exact solutions** can only be found using a **brute force** approach: **enumerate and evaluate all possible solutions** (usually, exponential or worse)
- **Reducibility**: any NP-complete problem can be transformed into another NP-complete problem (in polynomial time)
- Examples of NP-complete problems: TSP, knapsack, SAT, graph coloring, SubsetSum, Hamiltonian Path…

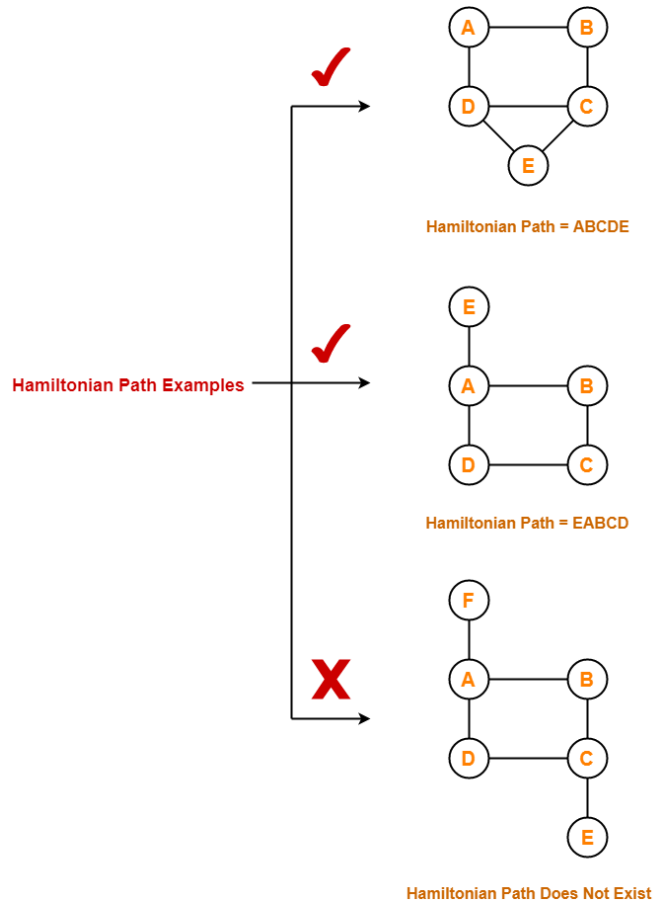


NP-C: Hamiltonian Path Problem (HPP)

- A path on a graph is Hamiltonian if it **starts at a specific vertex**, **visits every vertex exactly once**, and **returns to the first vertex**

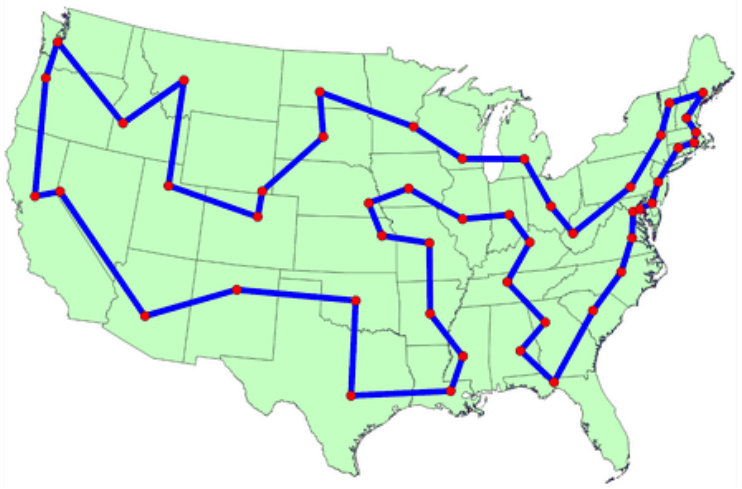
- **HPP: decide if a graph contains a Hamiltonian path**

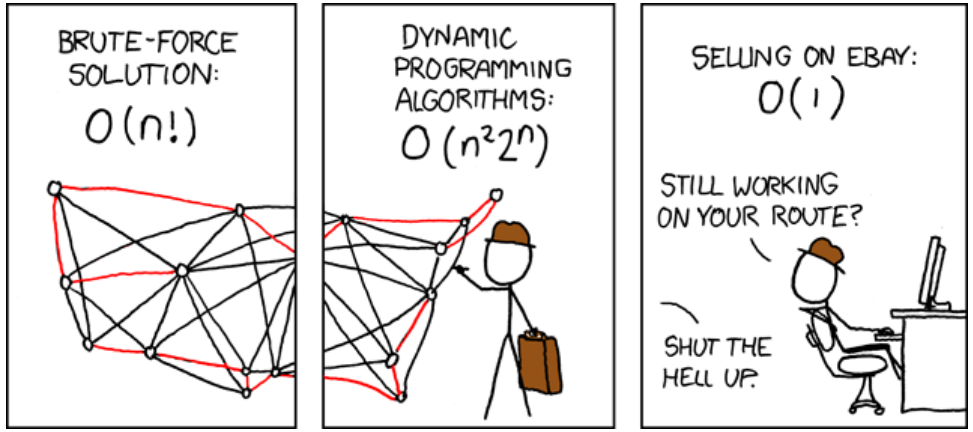
- **NP-complete**: no efficient algorithm is known, we must **enumerate all possible paths**
  - Alas, the number of paths is **factorial** in the number of vertices, that is, $\mathcal{O}(n!)$
  - **Impractical** even for small instances on conventional computers



NP-C: Travelling Salesman Problem (TSP)

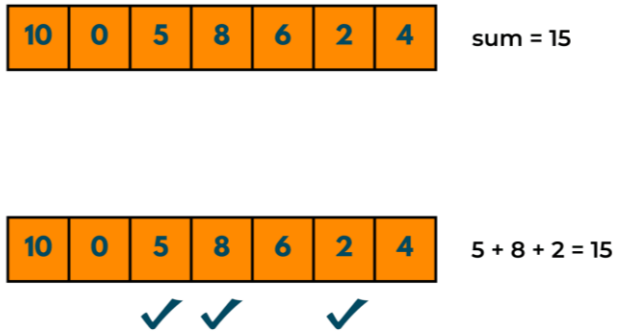
- The salesman must **travel to all cities exactly once** before returning home
- **The distance** between neighbouring cities **is given**, and it is assumed to be **identical in both directions**
- Goal: **minimize the total distance** to be travelled
- The **decision variant** of TSP is NP-Complete





NP-C: SubsetSum

- Assume a **sequence of numbers** B and a **value** v
- Find the subset of B such that the **sum** of the numbers equals v
- The decision variant («is there a subset…») is NP-complete



One interesting characteristics of NP problems

- To *find* a solution, in the worst case, **we must test all possible cases**, which often requires to explore a **huge solution space**

- To *check* whether a solution is valid (or not) is **much easier and faster**: we just **evaluate the solution** (in polynomial time)
  - E.g., in the TSP we just sum the distances between the cities; in Subset Sum we add the numers in the solution, etc.

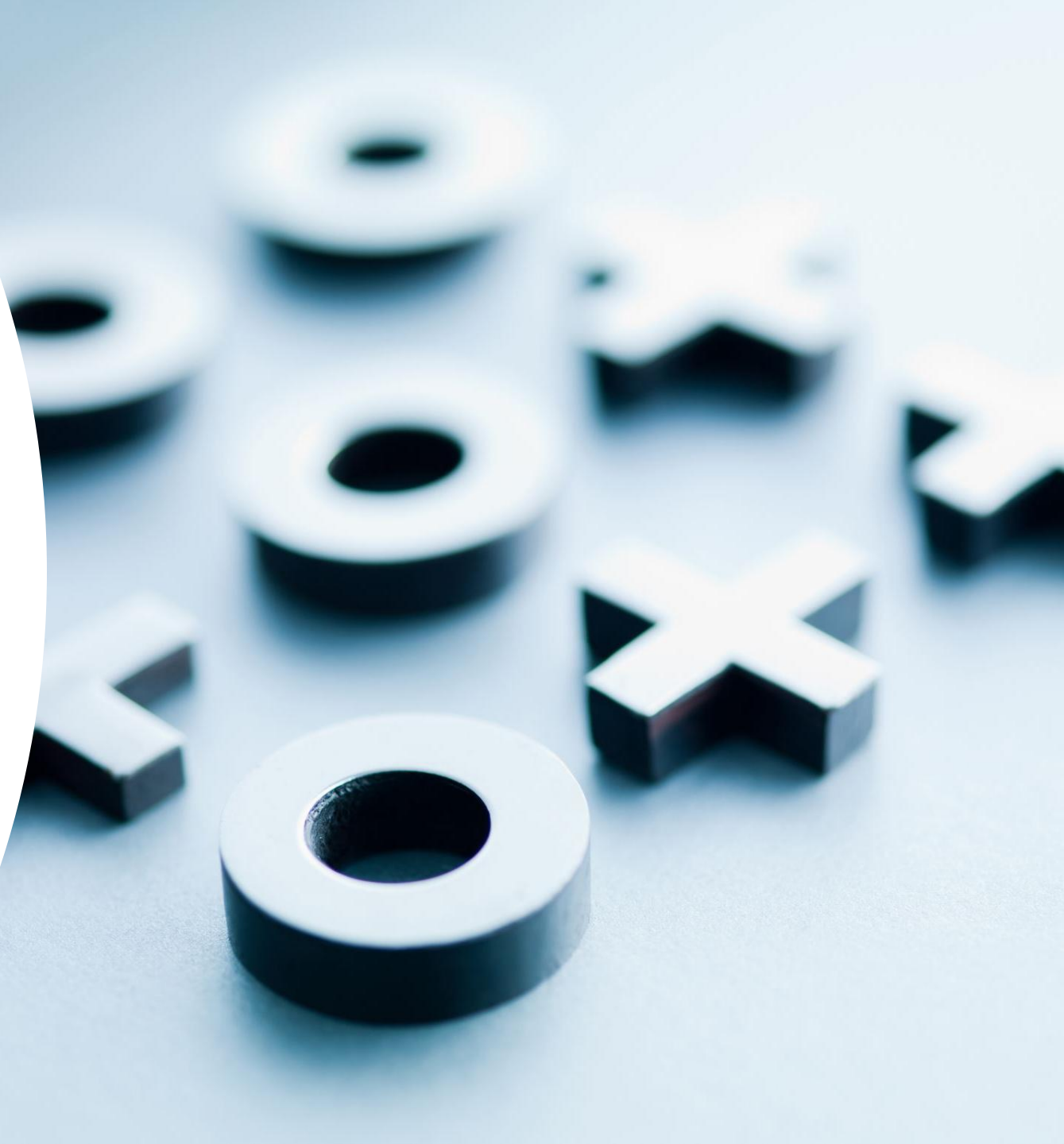
- Thus, an algorithm that solves an NP problem in polynomial time must be somehow able **to explore all possibile solutions at once** (!)

One key concept

NP does not mean «non polynomial complexity»

NP actually means **«polynomial on a *non-deterministic* Turing machine»**

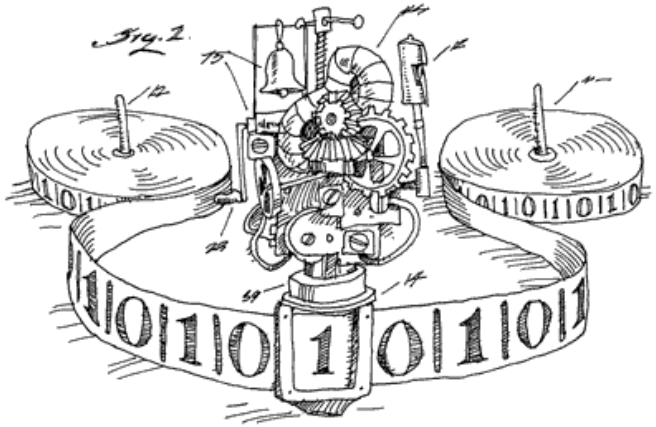
That is, a **special type of Turing machine**



Turing Machines [1/2]

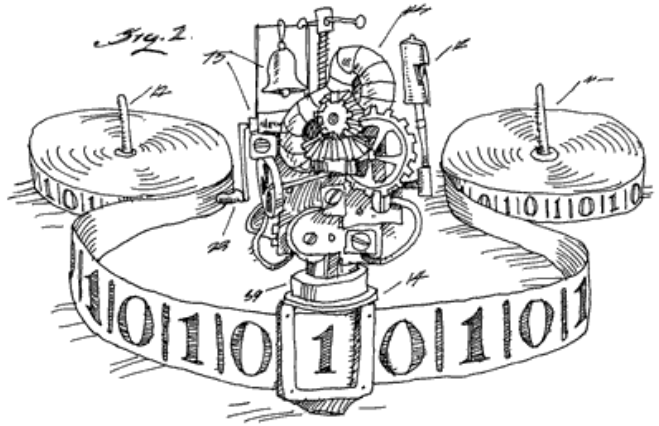
A (deterministic) Turing Machine (DTM) is composed of three parts:

- **A tape**, organized in **cells**, that can carry **symbols** (from a finite alphabet)
- **A head** over the tape. The head can **read** or **change** the current symbol on the tape, and can **move the tape** one cell to the left or right
- **A state**, selected from a finite set of states



Turing Machines [2/2]

- At each step, the head **reads the symbol** on the tape
- Based on the **read symbol** and the **current state**, the DTM can **write a symbol** in that cell and can **move the head** (← or →) or **halts computation**
- **The choice** about what to write on the tape and whether to move or halt is based on a **finite table** that specifies the action for **all pairs** (state, symbol)

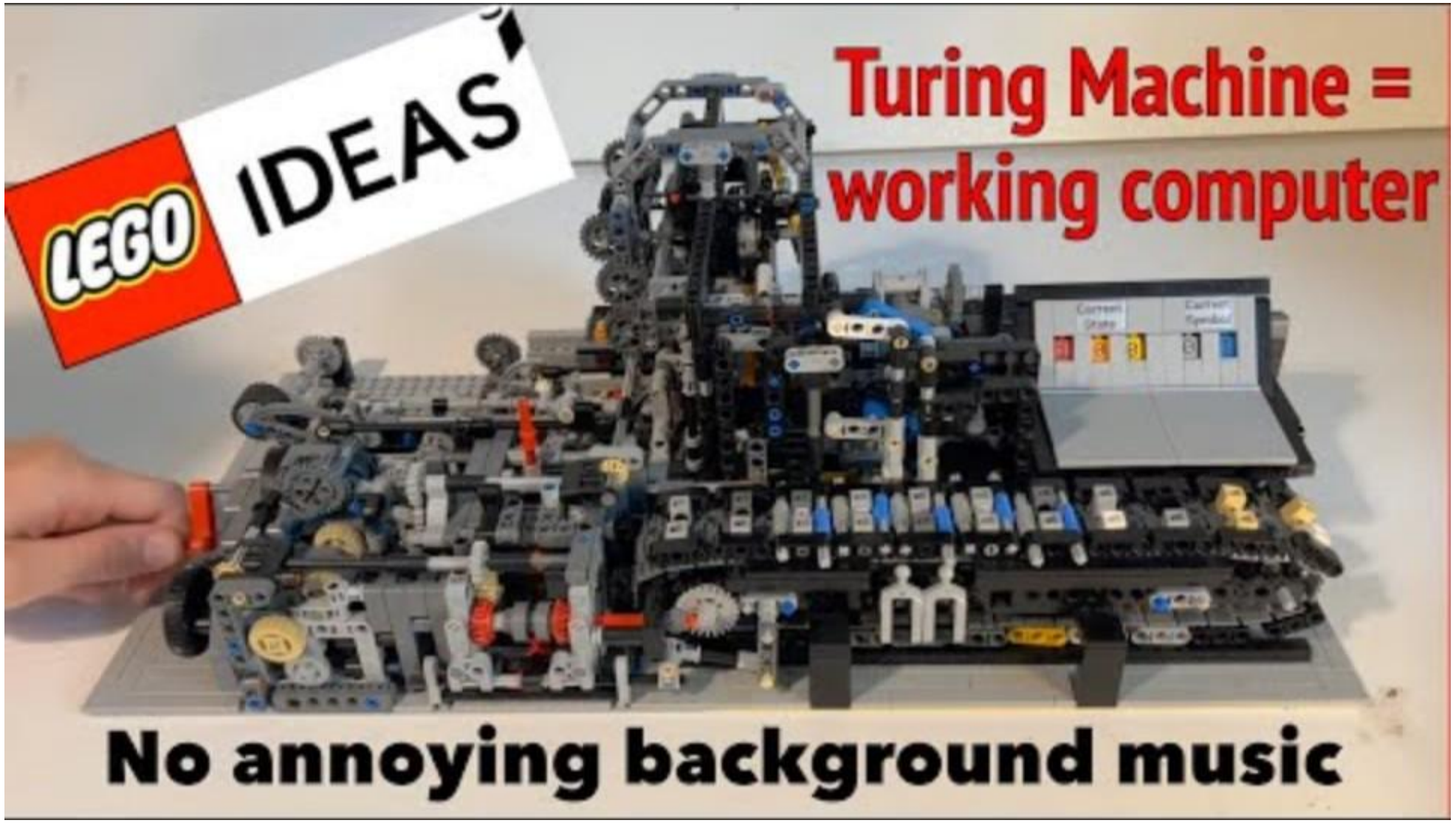


Symbols

States δ	0	1	B
q_0	$(q_0, 0, R)$	$(q_0, 1, R)$	(a_0, B, L)
q_1	–	–	–
a_0	(s_0, B, L)	(s_1, B, L)	(q_1, B, N)
a_1	(s_1, B, L)	(s_2, B, L)	–
a_2	(s_2, B, L)	(s_0, B, L)	–
s_0	(a_0, B, L)	(a_2, B, L)	(q_1, B, N)
s_1	(a_1, B, L)	(a_0, B, L)	–
s_2	(a_2, B, L)	(a_1, B, L)	–

← Example of programming table

LEGO Turing machine



Back to NP-complete problems…

- **Exact solutions** can only be found using a **brute force** approach: **enumerate and evaluate all possible solutions** (usually, exponential or worse)

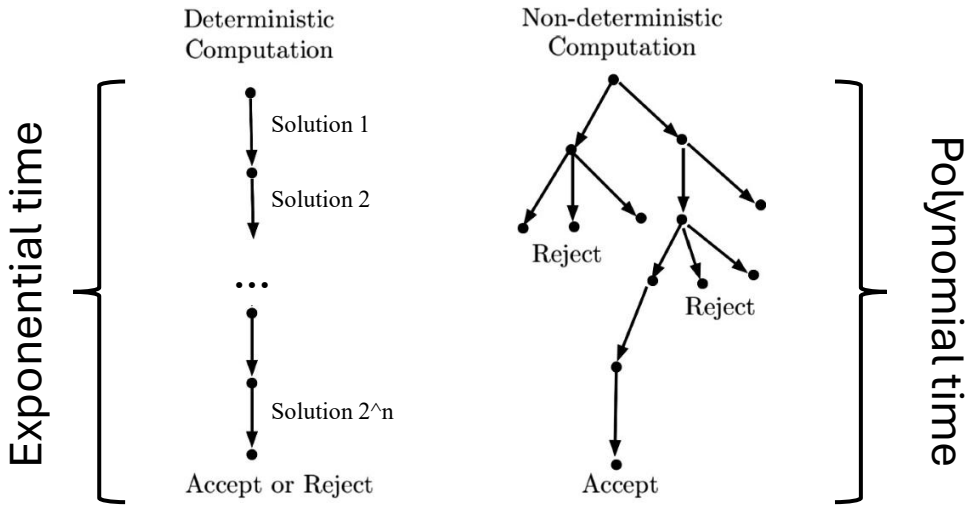
- Thus, on a DTM, we must **generate a large number of possible solutions** and **verify each one of them** (in polynomial time)

- Let us imagine if this was not the case…

Non-Deterministic Turing Machines (NDTM)

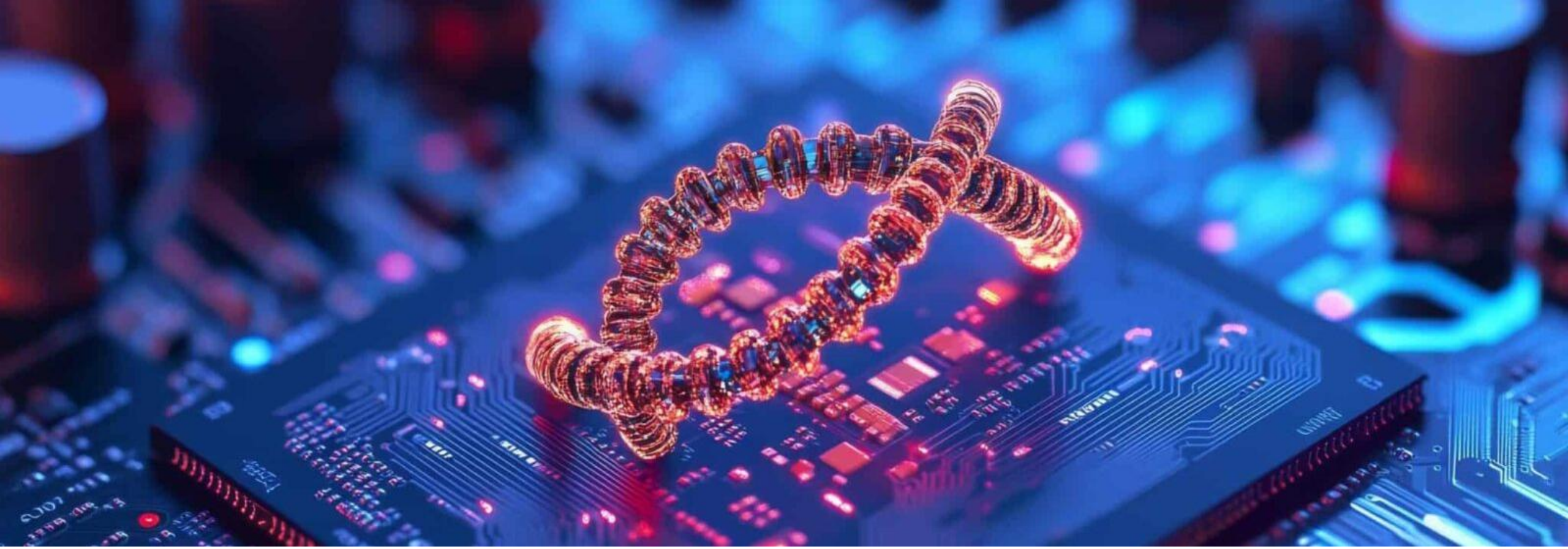
- To get a **solution of an NP problem in polynomial time** the machine must be able to **explore all possible solutions simultaneously**

- Or, as an alternative, we need an **«oracle»** module that gives us a solution



- In both cases we easily **check** the solution in **polynomial time**

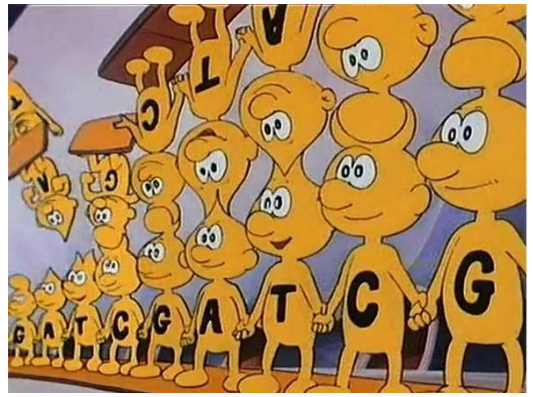
- Nice. HOW?



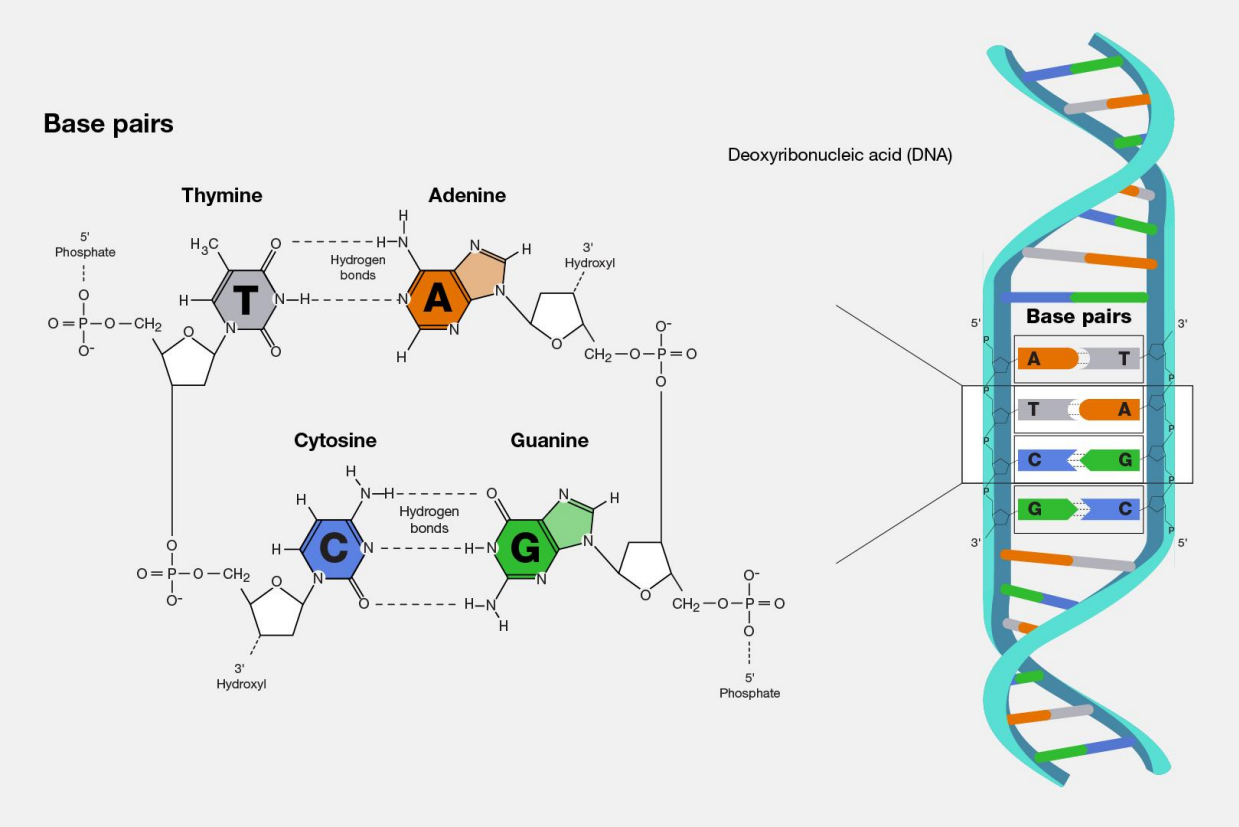
Computing with DNA

DNA computing is based on the idea of **exploiting spontaneous DNA self-assembly** processes to perform computation

DNA

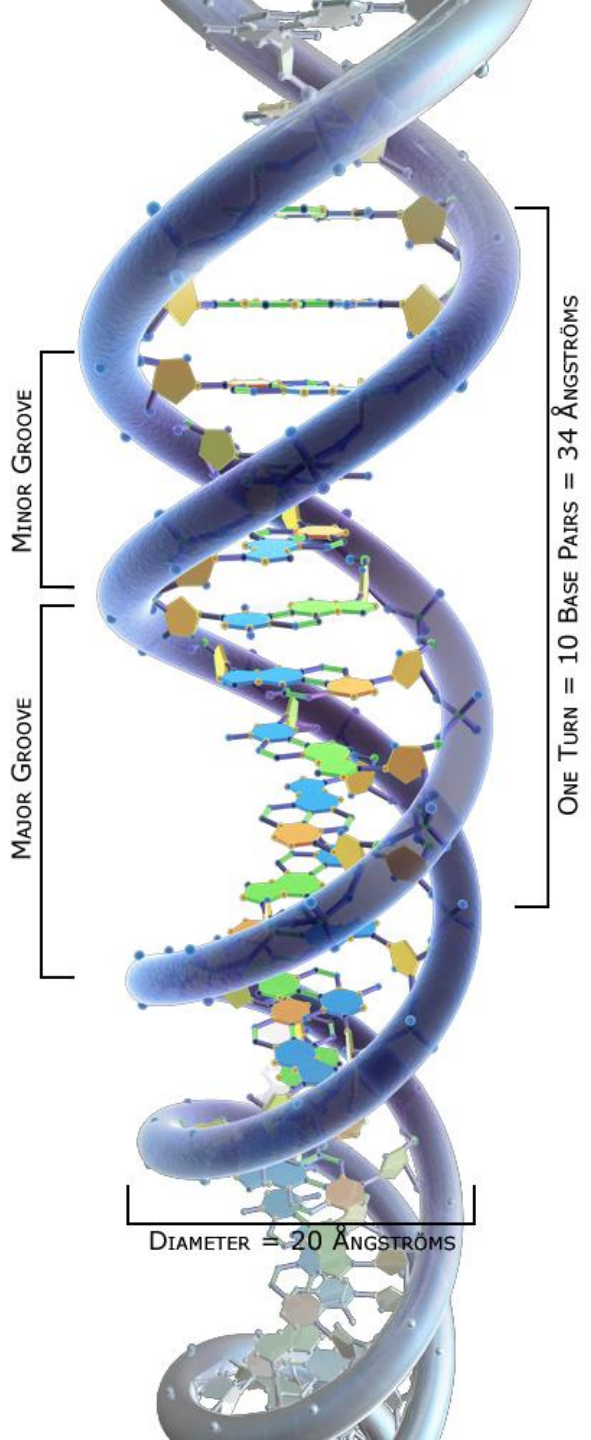


- DNA is made of **two linked strands** that wind around each other
- Each strand has a backbone made **of alternating sugar** (deoxyribose) **and phosphate groups**
- Attached to each sugar is **one of four "bases"**: adenine (A), cytosine (C), guanine (G) and thymine (T)
- **A base pair** consists of **two complementary DNA nucleotides** (A-T, C-G)



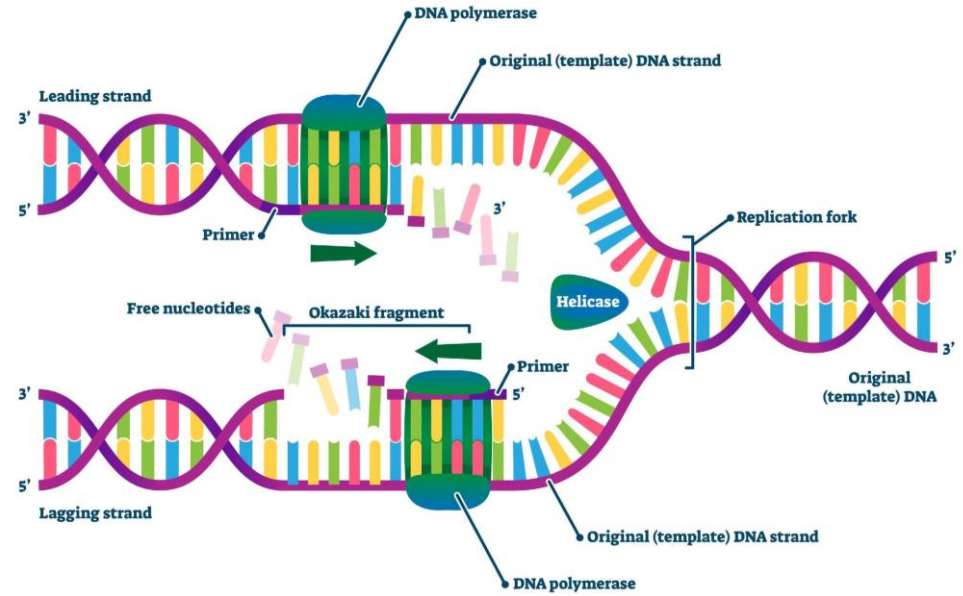
Structural properties of DNA

- DNA is very stable, regular, a rigid crystal: desirable characteristics to **store information** and **build artifacts**
- DNA has a minor and a major groove
- The double helyx makes a complete turn **every 10 base pairs**
- It can be arranged in **regular lattices**
- The diameter is **20 Ångstroms** (2 nm)



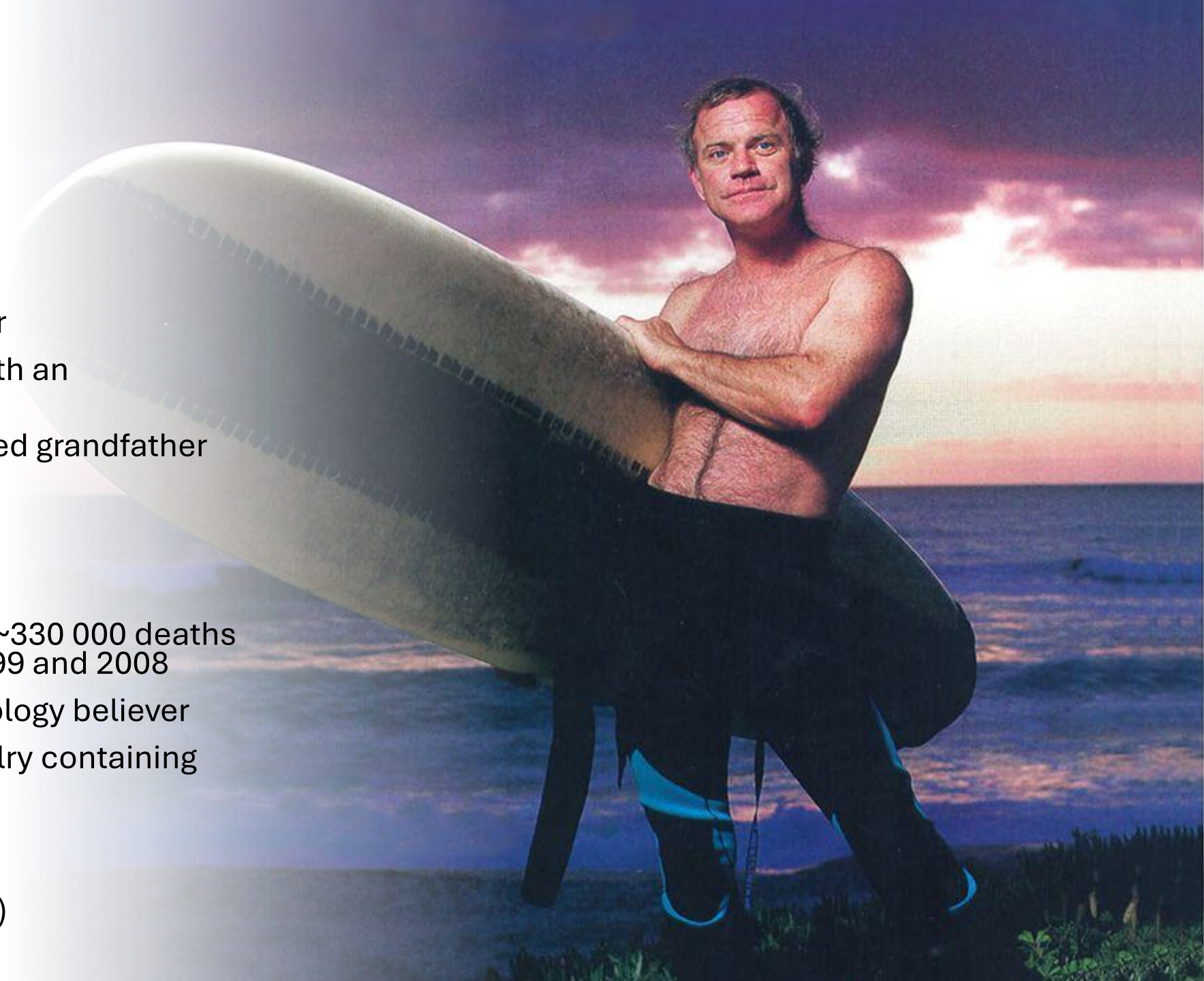
Polymerases

- **DNA polymerases** are **enzymes** that are responsible for **DNA duplication**
- They **read** the code from one strand and form the **complementary sequence** on the other strand by **assembling free nucleotides**
- Polymerase does not start replication on its own but **requires a «primer» sequence**
- Interesting fact: it is **extremely well conserved** across all organisms, it seems irreplaceable



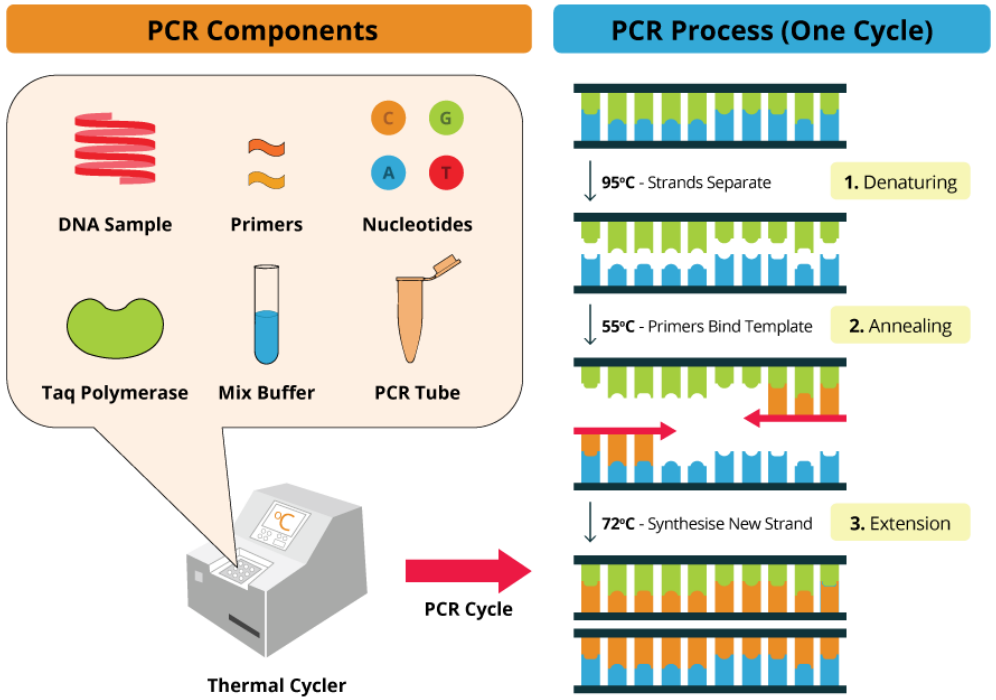
Kary Mullis

- Surfer and guitar player
- Science fiction writer
- LSD producer and fond user
- The first man to ever talk with an alien fluorescent raccoon
- Had a beer with his deceased grandfather
- Former bakery owner
- Climate change denier
- Ozone layer denier
- HIV denier, responsible for ~330 000 deaths in South Africa between 1999 and 2008
- Space traveler and yet astrology believer
- Unauthorized seller of jewelry containing Marylin Monroe's DNA
- **Nobel prize in Chemistry**
- **Inventor of the PCR** (polymerase chain reaction)



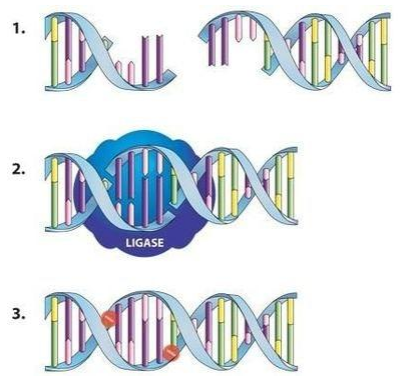
PCR

- **Polymerase Chain Reaction**
- **Amplifies a DNA signal**
- Process:
  - Put the DNA sample, nucleotides and polymerases in a test tube
  - Denaturate the DNA by heating the sample
  - Attach the primers to the region to be amplified
  - The polymerase will bind to the primers and extend the DNA region
  - Rinse and repeat 30x
  - The tube is now filled with the amplified signal marked by primers



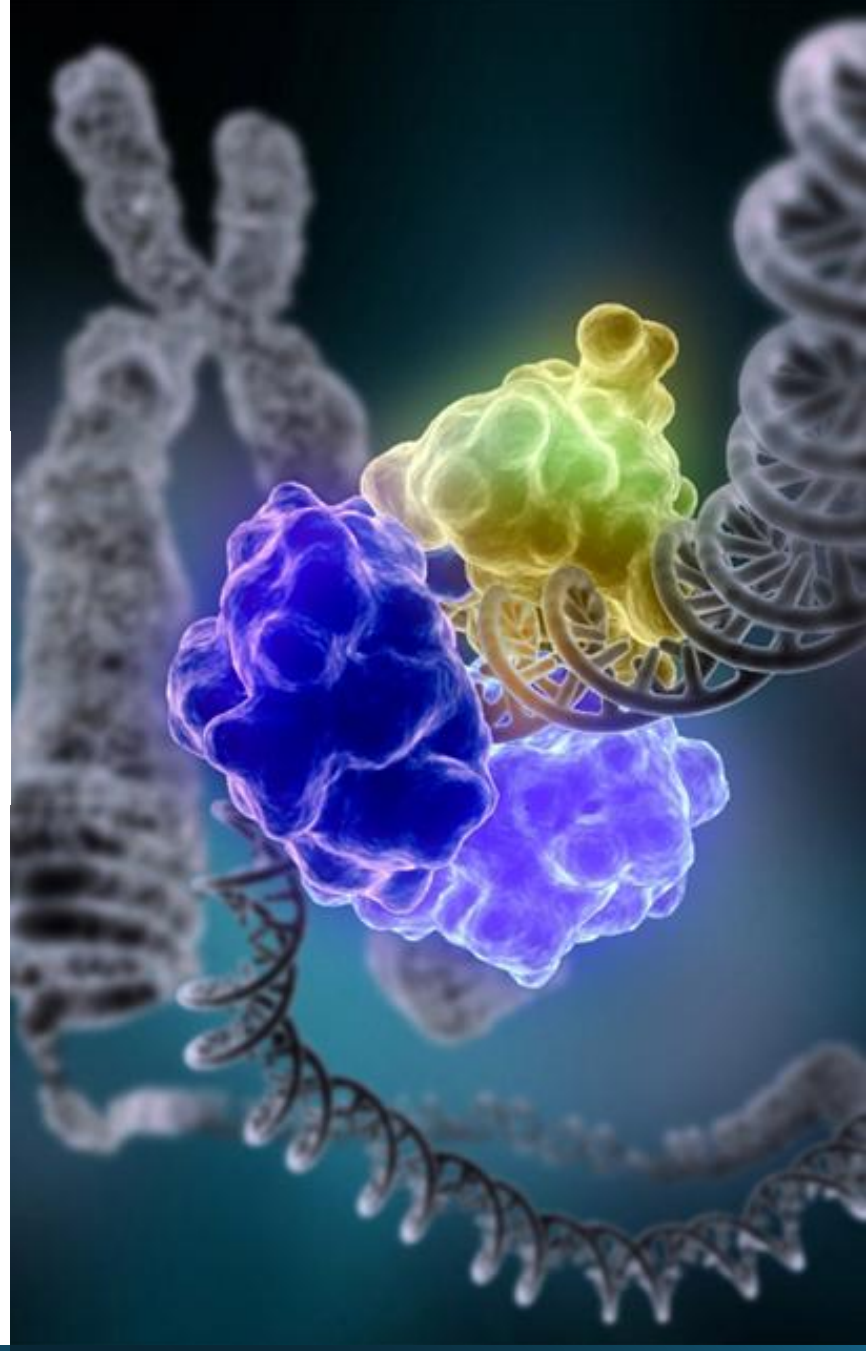
Ligase

- A Ligase is an enzyme that **catalyzes the joining** (ligation) of two molecules



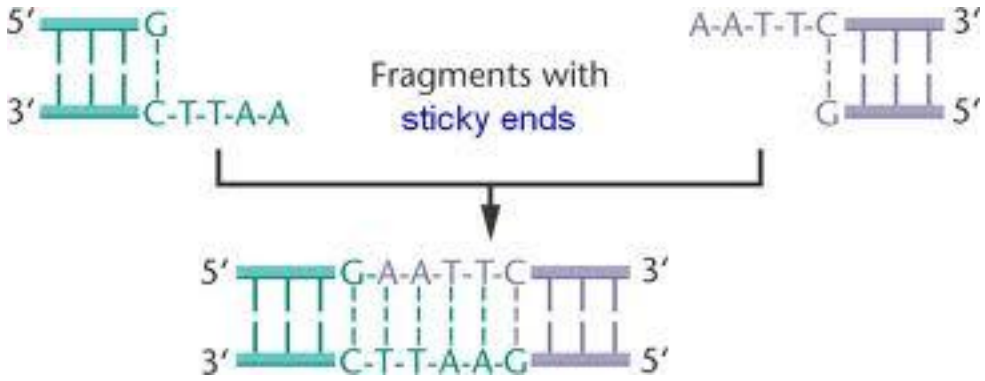
- **DNA ligase** can accelerate the join between **two DNA complementary fragments** by phorming phosphodiesteres bonds

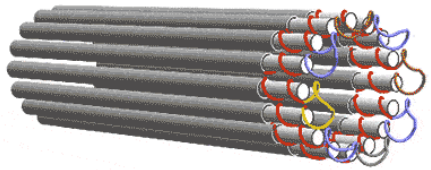
- Such complentary fragments are called **«sticky ends»...**



«Sticky ends»

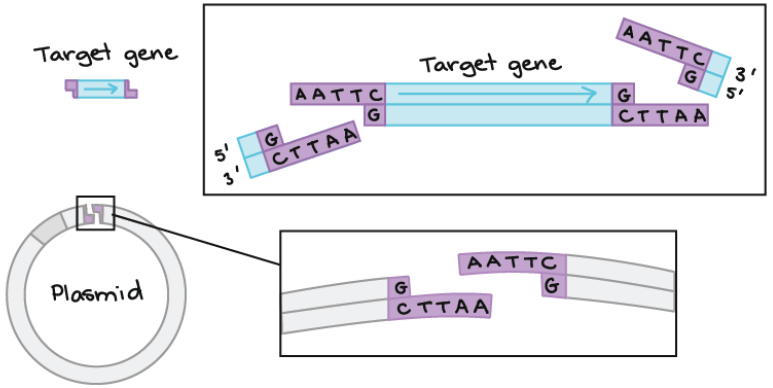
- If two DNA fragments expose **single strands fragments with complementary bases**, they tend to **attach to each other**
  - The process is accelerated by DNA ligases
- This process is known as **DNA self-assembly**
- Such exposed strands are called **sticky ends**





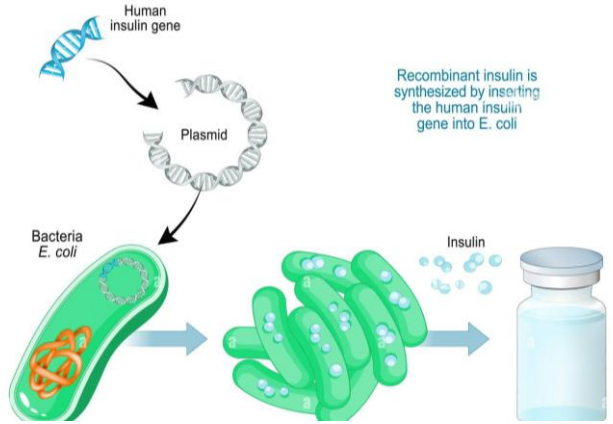
What can we do with sticky ends?

- For instance, we can take a plasmid (circular DNA code, typical of bacteria), **break it to expose sticky ends**, and **insert a new gene** decorated with **sticky ends** matching those of the plasmid



- Now that bacterium will also **express the new gene**

- This is how **insuline** is produced!



…or we could build a computer!

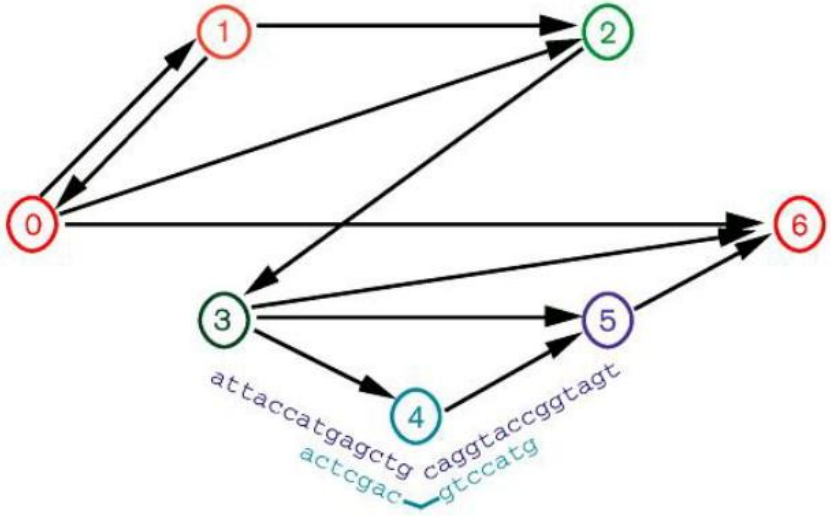
- **Leonard Adleman** (Turing award 2002), computer scientist, co-inventor of the **RSA encryption algorithm**
- He pioneered the **use of DNA to perform computation**
- His computer was used in 1994 to **solve the HPP!** [Adleman, Science 1994]

Adleman’s DNA computing device

- Adleman had the intuition of **encoding pieces of the HPP solution with DNA**
- The idea is that solutions of the HPP will **spontaneously self-assembly in the test tube**
- We can read the result after the self-assembly process is completed

DNA encoding in Adleman's computer

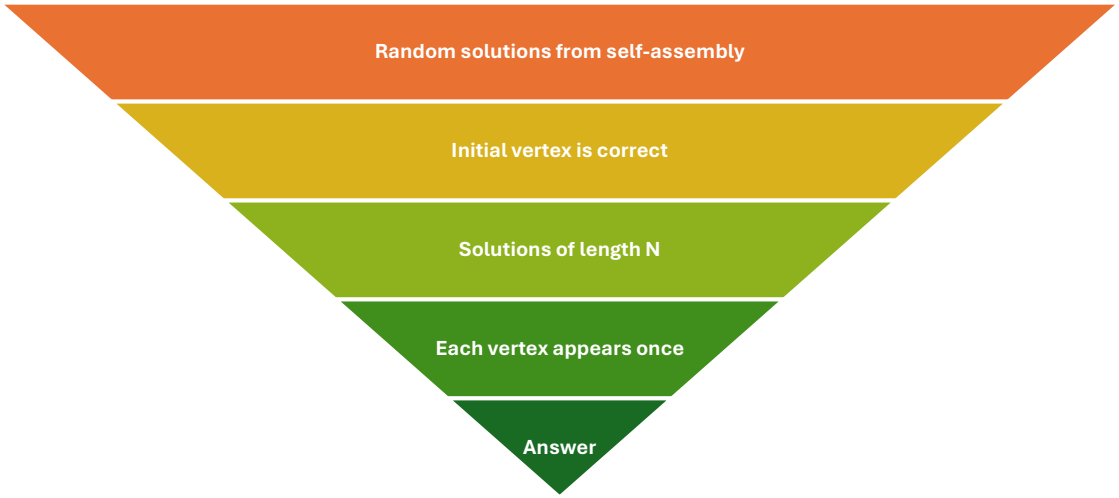
- Each **vertex of the HPP** is represented by a (unique) **20-bases DNA single strand**
  - Why so long? To diversify the encoding as much as possible to prevent wrong assemblies



- The **connection/edge** between the cities/vertices is encoded by a **10-bases strand**

- By putting **plenty of such strands** in a test tube we can exploit spontaneous assembly to **explore a massive amount of parallel solutions**

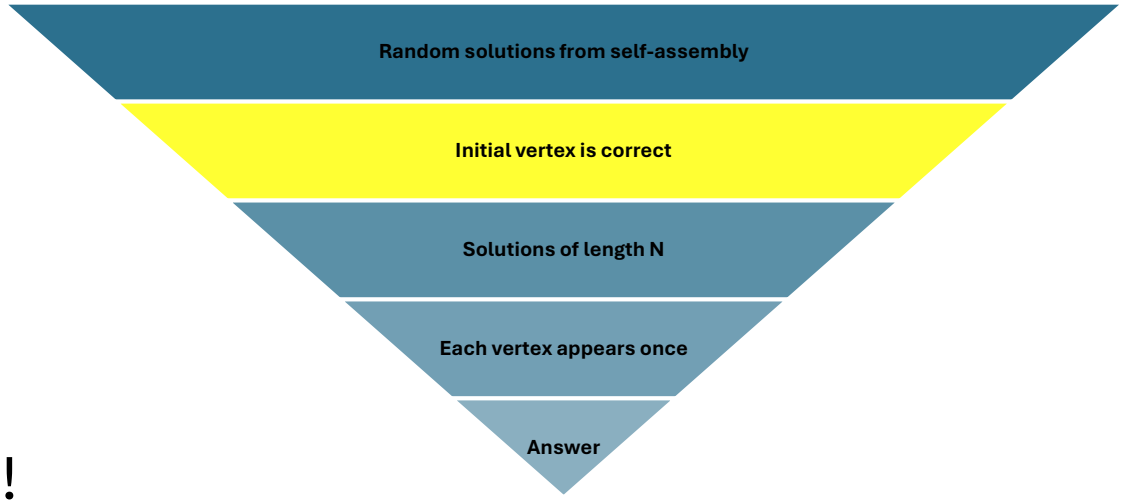
Identifying the solution



After we generated the candidate solutions in the test tube, **to detect whether there is an actual solution** to the HPP we need to perform the following steps:

1) Drop any residual single-strand DNA
2) Keep only those solutions that **start and end with the given vertex**
3) Keep only the **solutions of length N**, where N is the number of vertices
4) Keep only those solutions where **each vertex appears only once**
5) The remaining DNA double strand is the **solution** of the HPP!

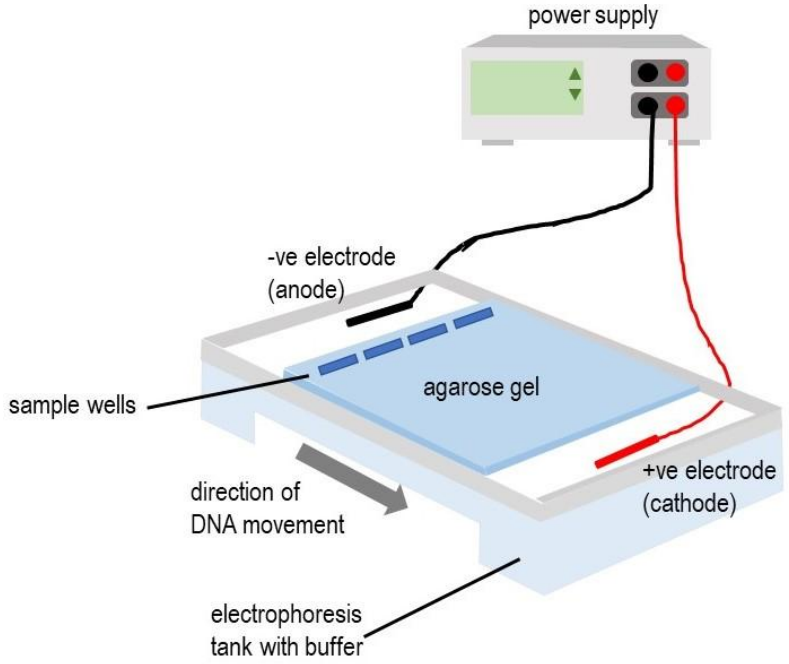
Correct initial vertex

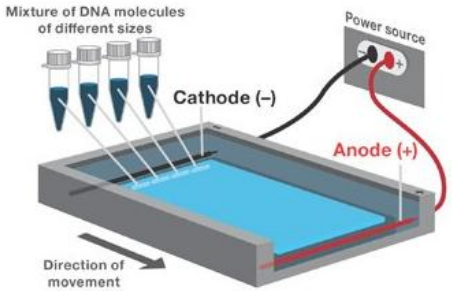


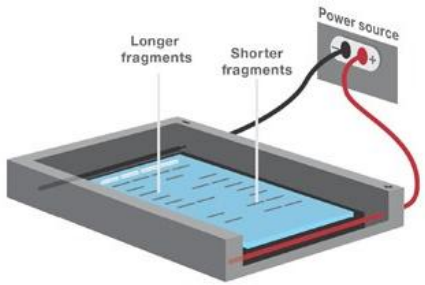
- We use PCR with **a primer that binds to the first vertex**: it will amplify only those sequences that start from the right «city»!
- Any other sequence will basically disappear from the results
- If we want a specific **final vertex**, we can use PCR and **another primer**

Checking the length with gel electrophoresis

- Gel electrophoresis **separates DNA strands according to their size**
- It is based on an **electrophoretic chamber filled with agarose gel**
- **DNA is attracted** by the positive catode
- The gel acts as a sieve: **smaller molecules travel longer distances**



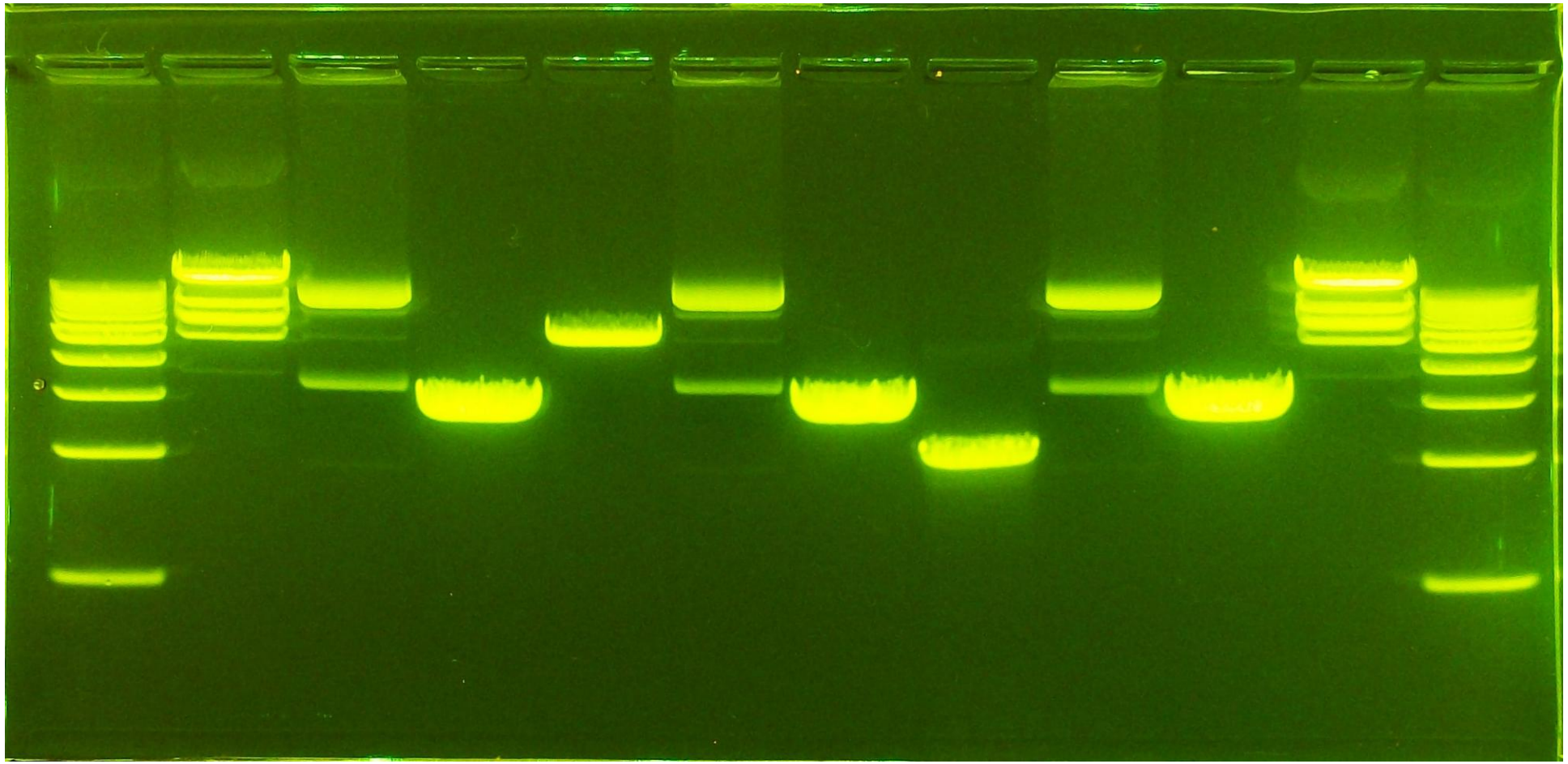




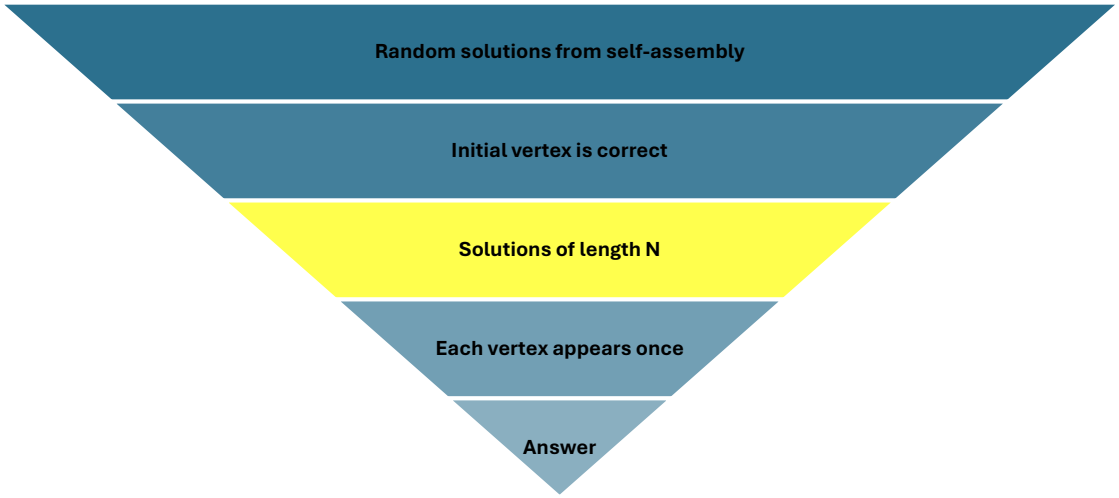
Example of gel electrophoresis experiment

← Experiments →

← Size / # bases →



Actual DNA experiment



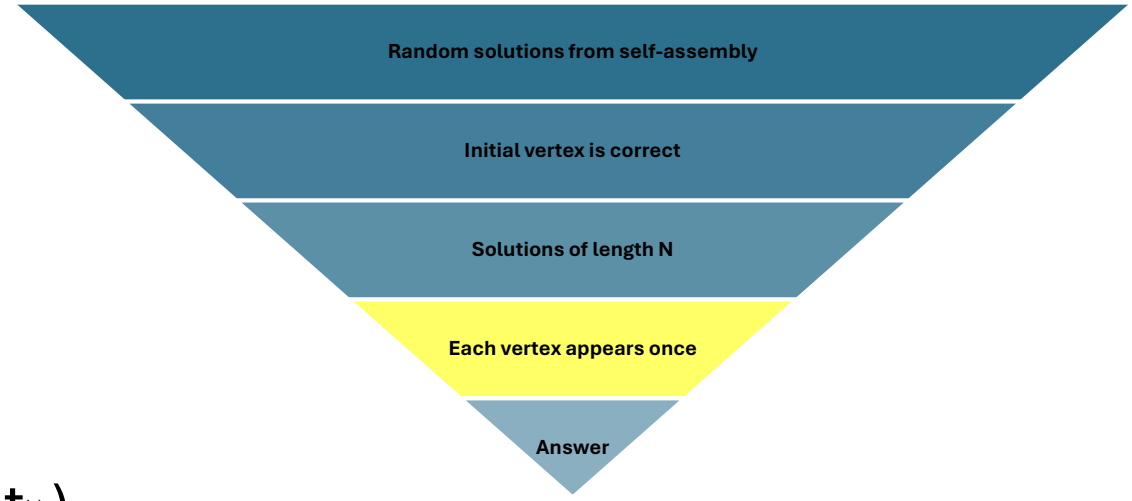
- From [Chang-Muk et al., Molecules Cells 1999]
- **18-mer strands** for vertices, **8 vertices, 14 edges**
- Solution must have exactly 18*8 = **144 bases**

M) Marker of 50 bases

1) Raw DNA material

2) DNA material + ligases

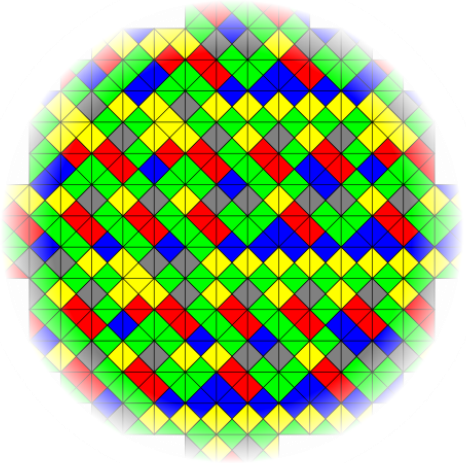
Each vertex appears once

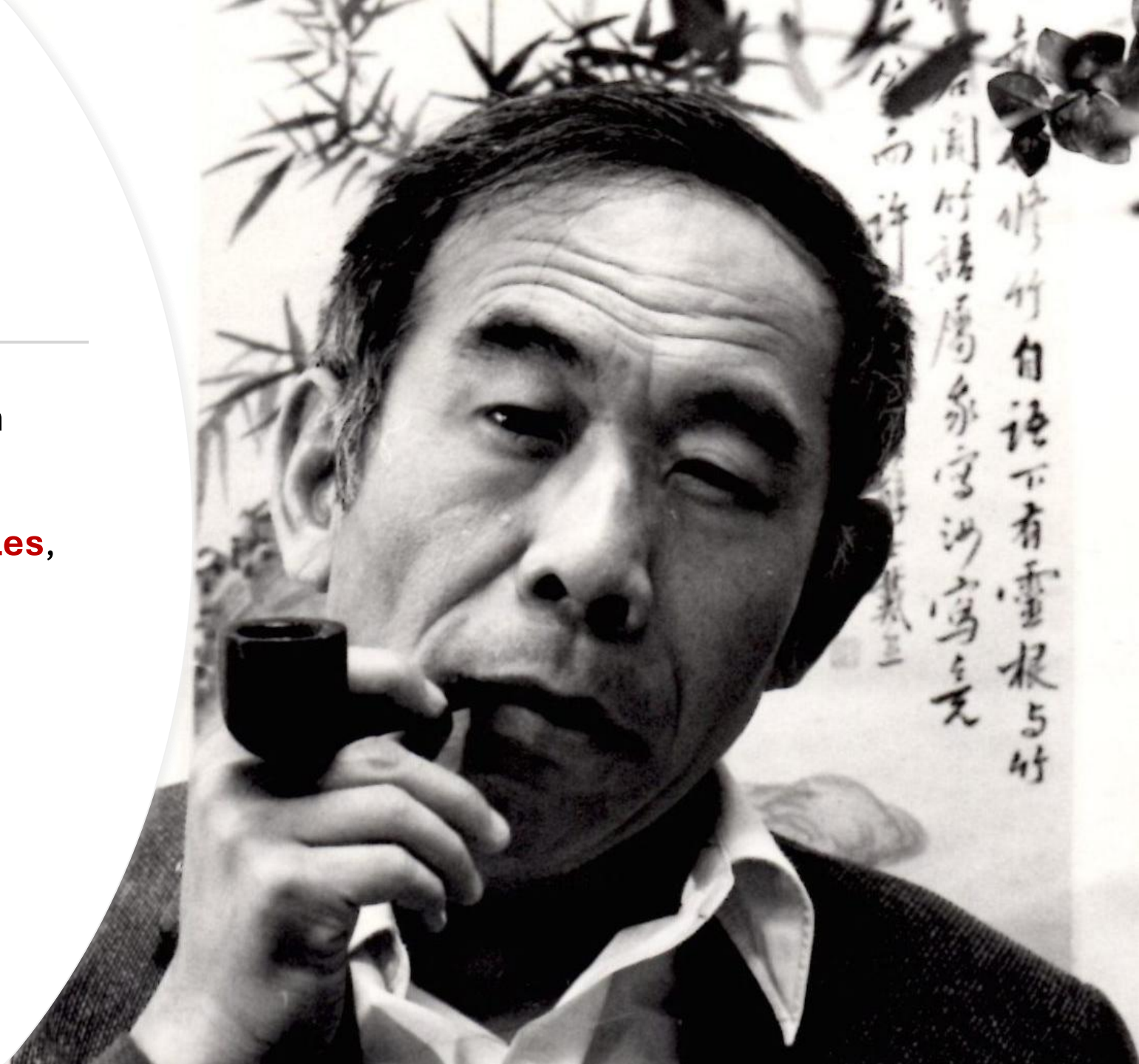


- Adleman used **affinity separation** («the most tedious part of the experiment»)
- He used DNA **«probe» molecules** that can attach to each city/vertex
- The probes were attached to **microscopic iron balls** (~1 micron)
- The idea was to attach the iron balls to each city, one by one, and **transfer the solutions to the next tube** by using magnets
- This way, only solutions that contain all cities can arrive to the final stage
- If the tube is not emtpy, the answer is YES

Hao Wang

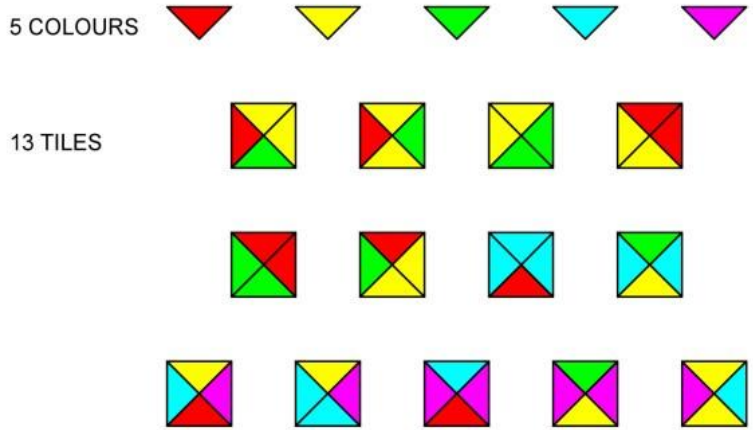
- **Logician**, philosopher, mathematician
- Most important contribution: **Wang Tiles**, **square** tiles with a **color** on each side



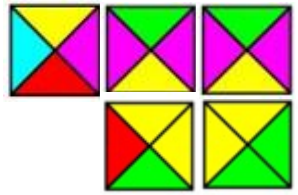


Wang Tiles (WTs)

- **Square tiles**
- Each tile **edge has one specific color**
- There is a fixed amount of **possible colors**
- Tiles are **arranged side by side in a grid**
- **Tiles cannot overlap**
- Each edge **must match the edge of adjacent tiles**
- Tiles have **fixed orientation**
  (i.e., are not rotated nor reflected)



Example:

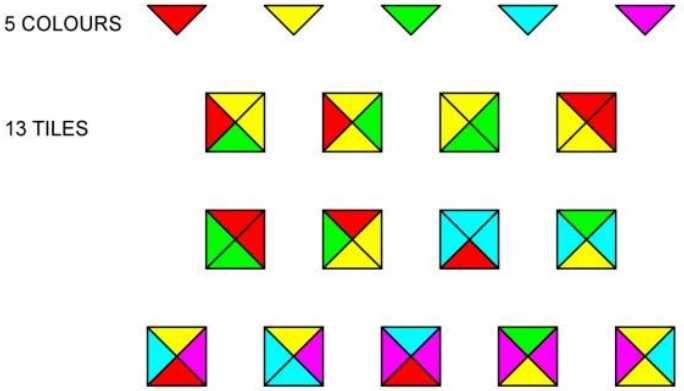


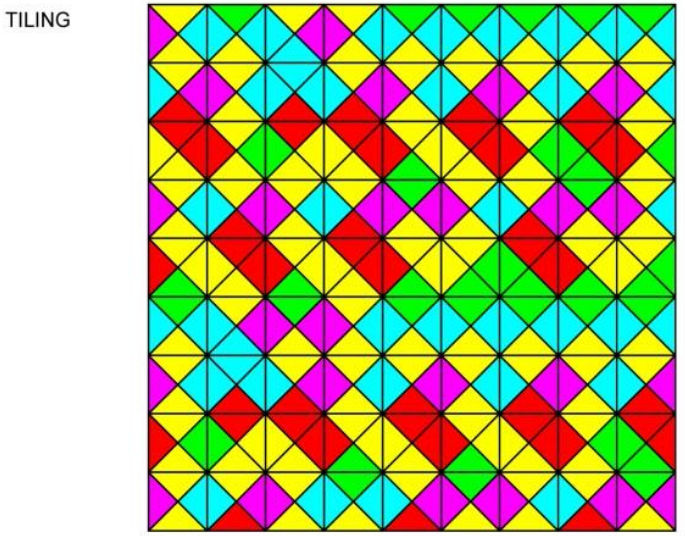
One surprising result of WTs: aperiodicity

- WTs **do not necessarily produce periodic patterns** and tilings

- It was also showed that WTs **do not always tile the plane**

- Actually, the result is even **stronger than that**…





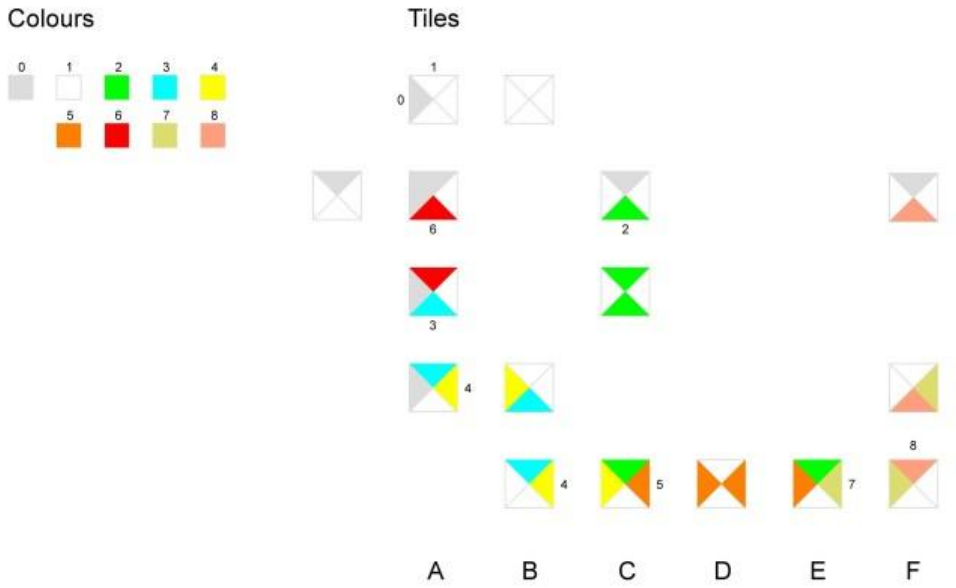
Computational power of WTs

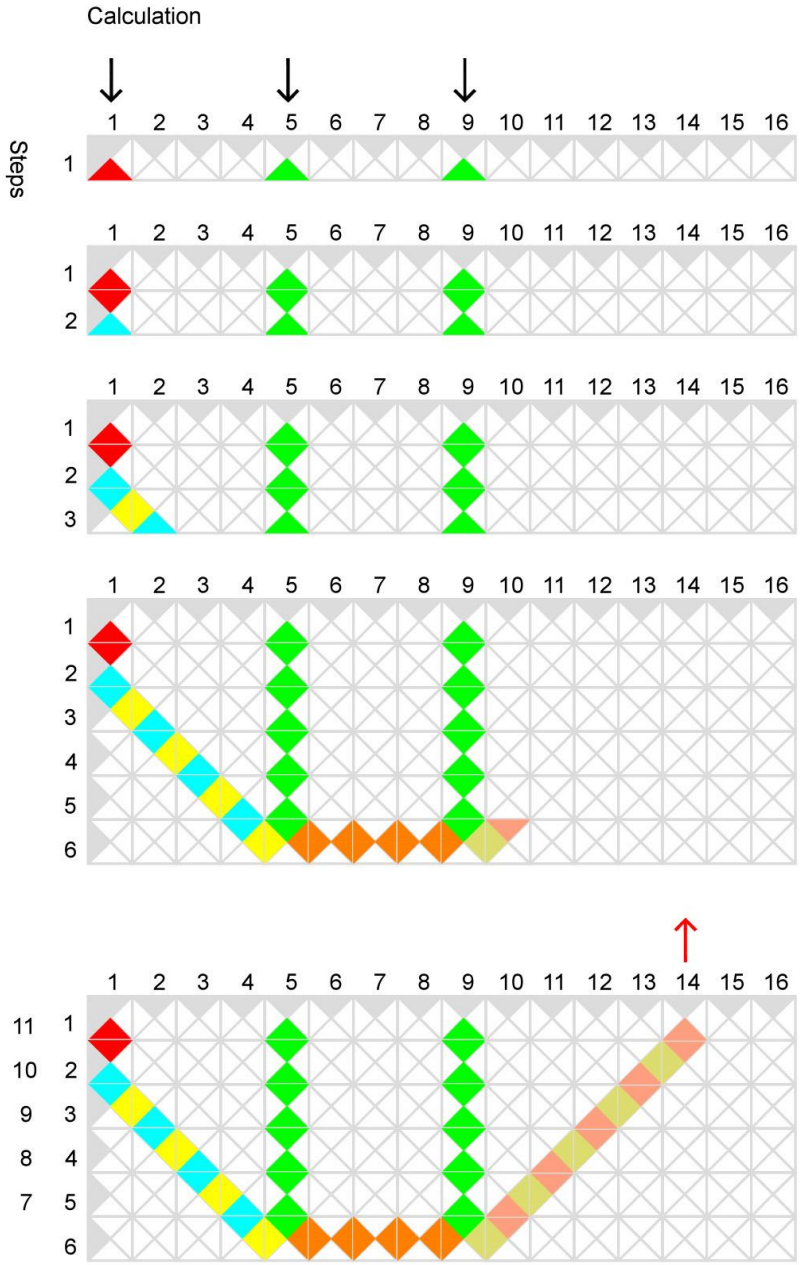
- **There can be no algorithm to decide whether WTs can tile the plane**
- It is possible to **translate a WT into a Turing machine that halts iif the tile set can tile the plane** → this is clearly impossible due to the **undecidability of the halting problem**
- As a matter of fact, **WT can simulate Turing machines**
- Thus, we can **create tiles that perform arbitrary computation!**

Summing two natural numbers with WTs

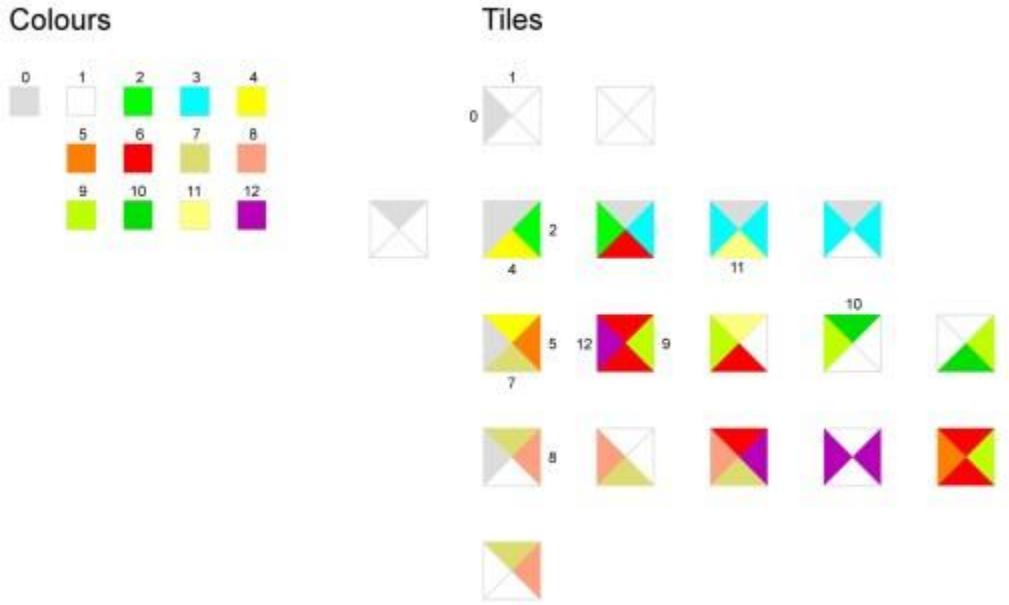
We use green tiles to «set» the numbers to be summed

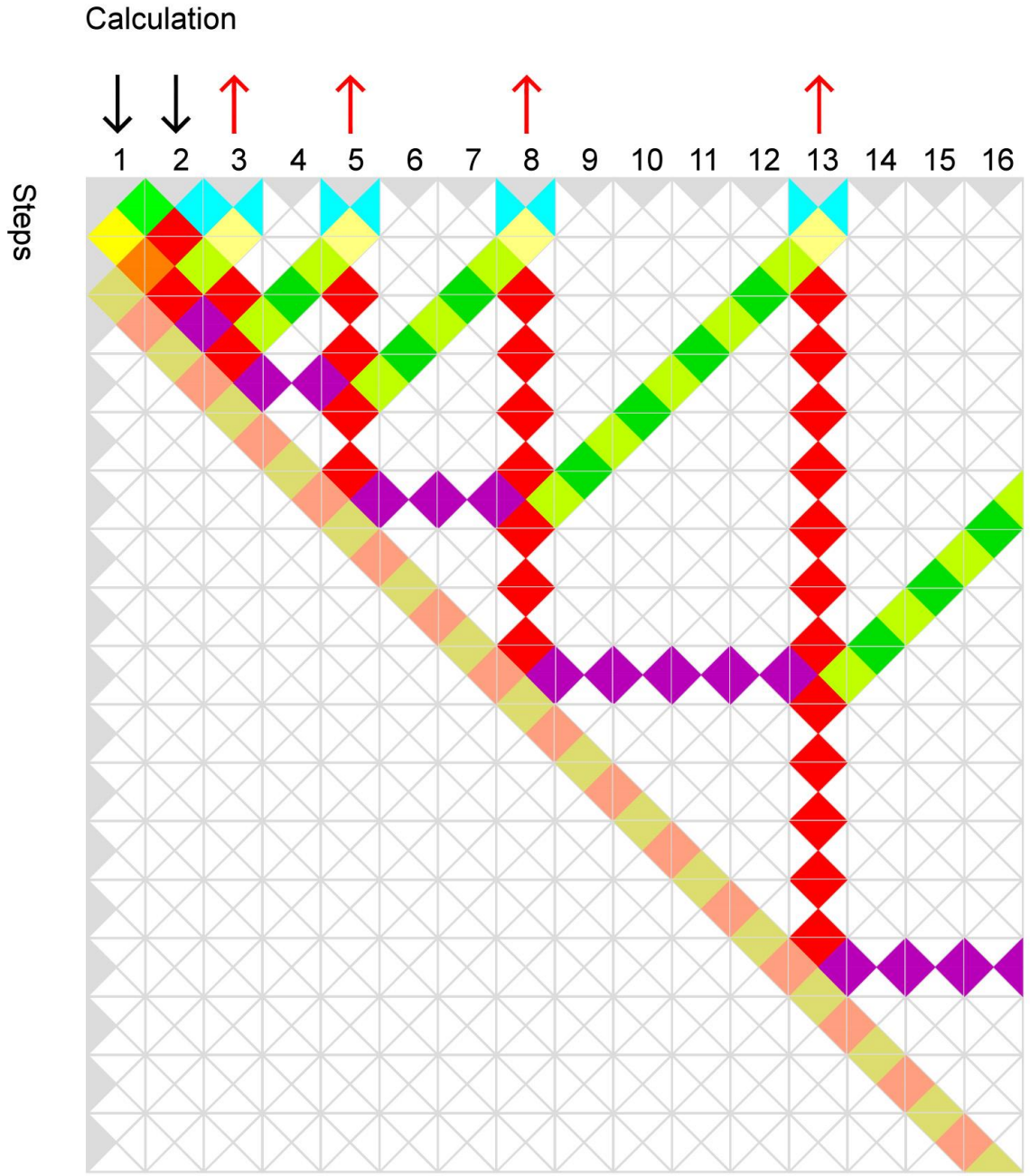
Computation starts from the red tile





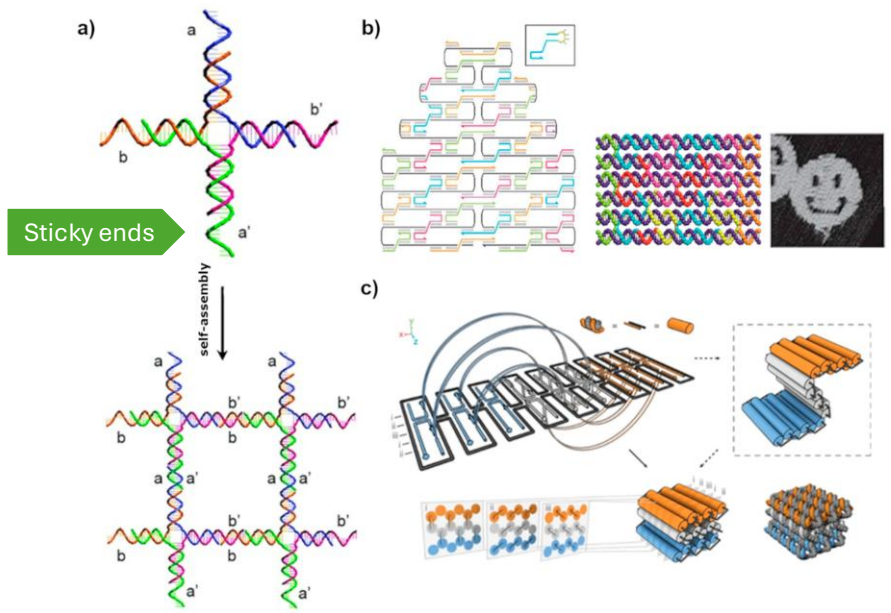
Another example: Fibonacci





DNA WTs

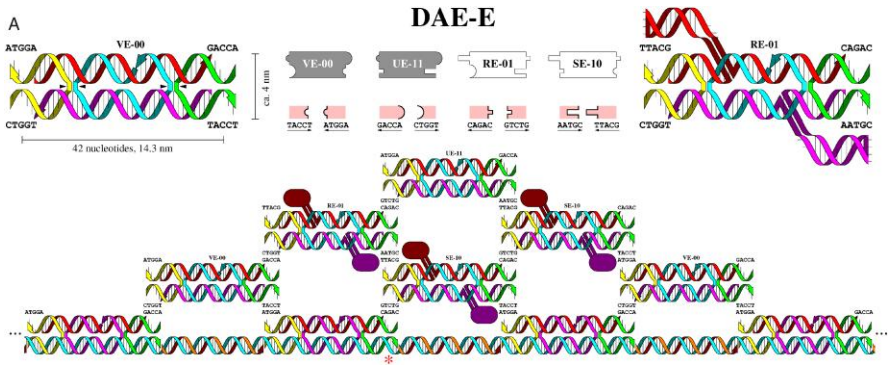
- We can create **DNA WTs with sticky ends**
- **By programming the DNA sequences** on the sticky ends, we can **control the assembly process**
- We can **build all sorts of «images»** and DNA structures on a nano-scale
- Scheme from [Hakker et al., J Biochem Struct Dynam 2013]



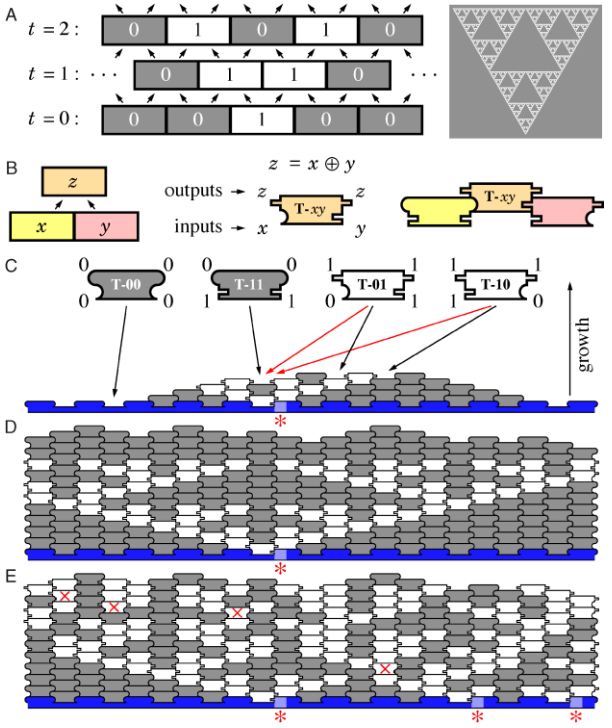
Algorithmic self-assembly of DNA Sierpinsky triangles

WT-based computation

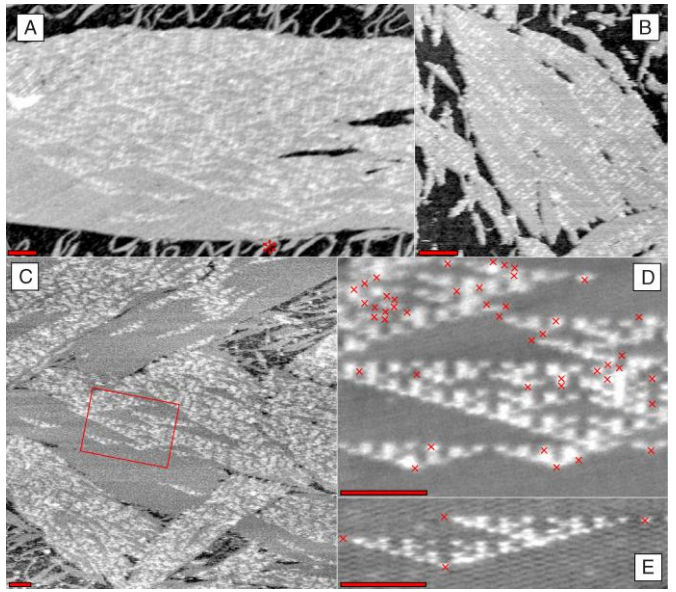
[Rothemund,..., Winfree, PLOS Biol, 2003]



Four tiles are sufficient to implement the rules! →

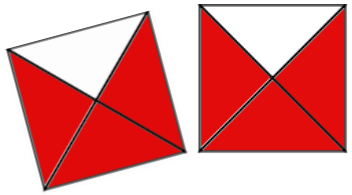


Atomic force microscopy reveals the Sierpinsky triangles →

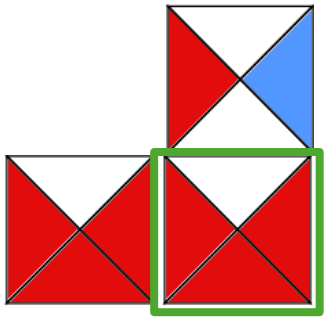


Implementing a NDTM with DNA WTs

- Consider again **DNA-based Wang tiles**
- Imagine that you **choose a temperature** for the experiment such that the binding of WTs is **stable only if at least two sticky ends are attached**

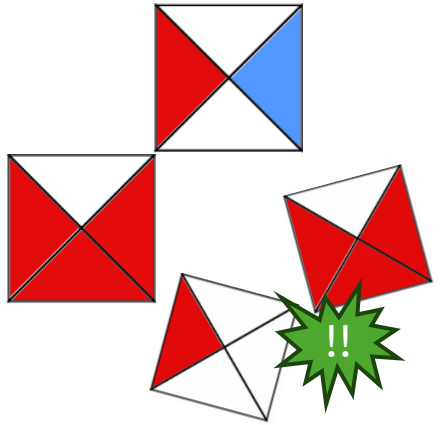


Not stable



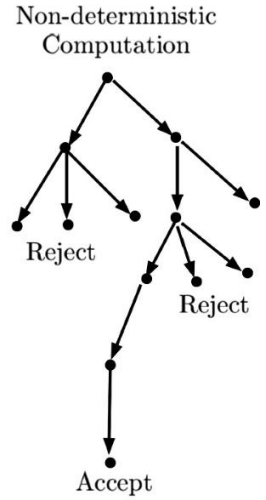
Stable

- Finally, design the sticky ends in such a way that **more than one tile type can attach** in the same position: the tiles will (stochastically) **fight for that position**



Non-determinism is the key

- We can create **billions of DNA WTs** and let them «run» in the test tube
- Thanks to non-determinism in the attachment, **we can explore multiple different computations at once**
- It can be shown that, with the right concentration, the probability of exploring **all possible computations** becomes close to 1
- That is exactly what we needed to implement a NDTM! One of the tile structures could be the **solution of our problem**



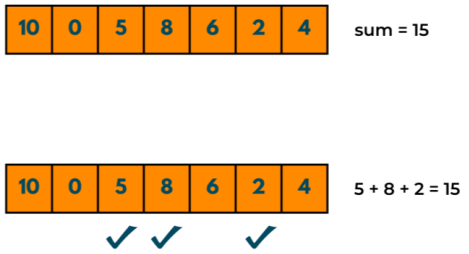
Some technical issues: input and output

- We need to provide an **arbitrary input** to the DNA system: we call that a **seed**

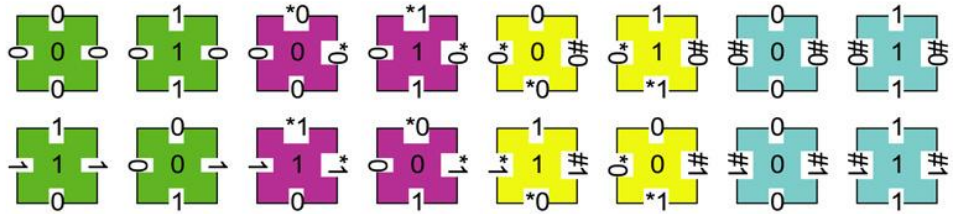
- We also need some sort of **«final tile»** that will be attached only **in the case of successful execution**
  - Possibly, engineered in such a way that we can **easily probe it**
  - One option: a tile decorated with iron or **gold nanoparticles**

One example: SubsetSum

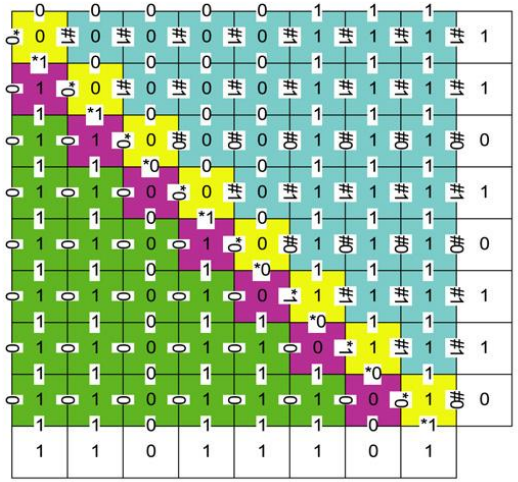
- Given a **sequence of numbers** B and a **value** v, find the subset of B such that the **sum** of the numbers equals v
- NP-complete problem
- **Brun solved SubsetSum with a NDTM based on DNA WTs** [Brun, TCS 2008]
- Five groups of tiles (56 DNA tiles in total):
  1) 16 tiles solving numbers **subtraction**, denoted by T_{-}
  2) 4 tiles implementing the **identity** function, denoted by $\mathbb{S}_{X}$
  3) 20 tiles implementing **nondetermistic guesses**, denoted by $T_{?}$
  4) 7 tiles are used to create the seed Γ_{SS}
  5) 9 tiles calculate the **final decision**, denoted by $T_{\checkmark}$
- The **seed** is a **L-shaped DNA crystal** with the binary representations of the **value** v (bottom) and the **input numbers** in B (right-most column)



Subtraction tiles T_{-}

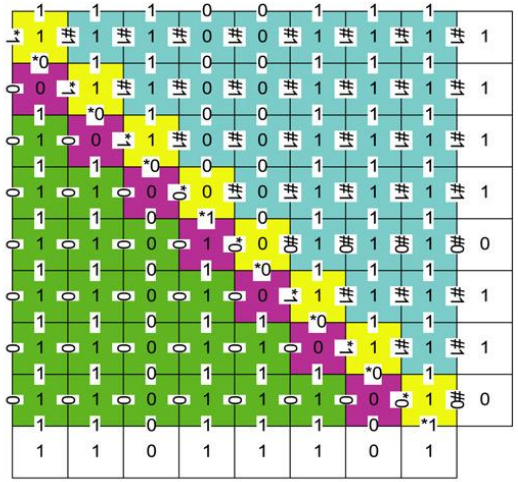


- The problem can be restated as: which subset of values should be **subtracted** from v?
- We need to be able to subtract values



(a)

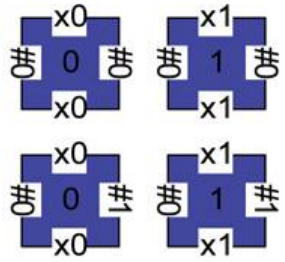
- In (a) we are calculating $221 - 214$
- 11011101 and 01101011, respectively
- The result is correctly 00000111=7



(b)

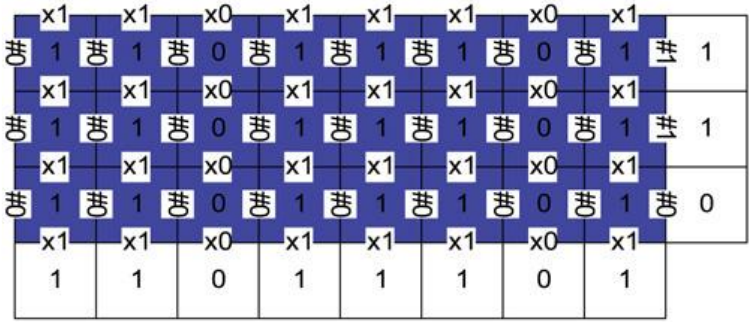
- In (b) we calculate 221-246
- 11011101 and 11110110, respectively
- **This computation fails**
- Failure is denoted by the yellow tile  in the top-left corner with west domain set to 1*
- More about failure later

Identity tiles $\mathbb{S}_X$

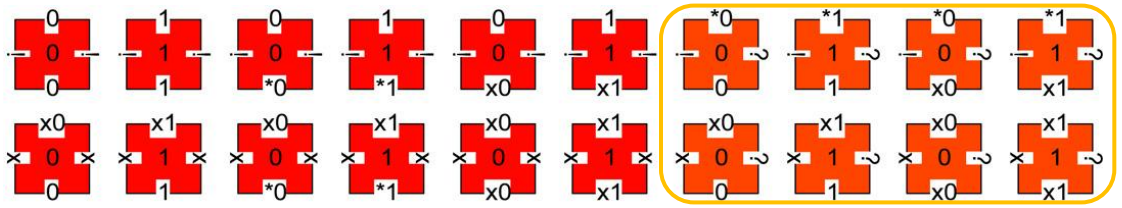


- These tiles just **copy the value** below to the row above
- These are used to **exclude one number** from the set (that is, not subtracted)

- In the example, the result of the previous iteration was 11011101 and that number is **copied** regardless of what is encoded in the seed on the right



Non-deterministic guess $T_?$

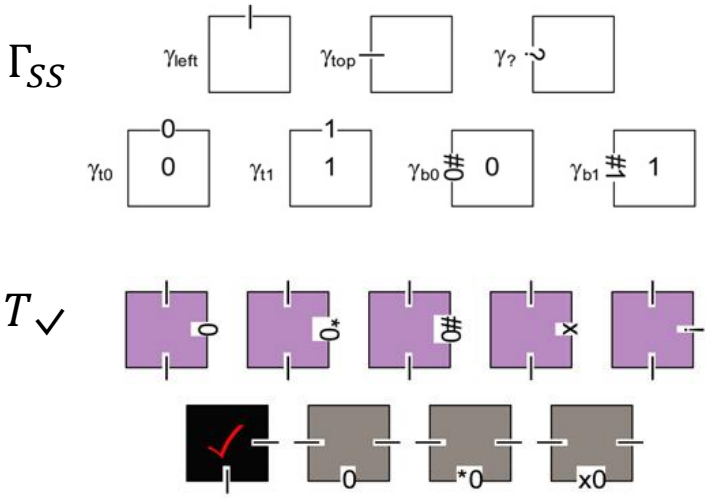


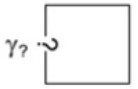
- These tiles implement the **non-deterministic choice** of which numbers in B should be considered for the solution

- Specifically, the tiles inside the orange box **do not have unique east/south binding domains**, thus will attach non-deterministically

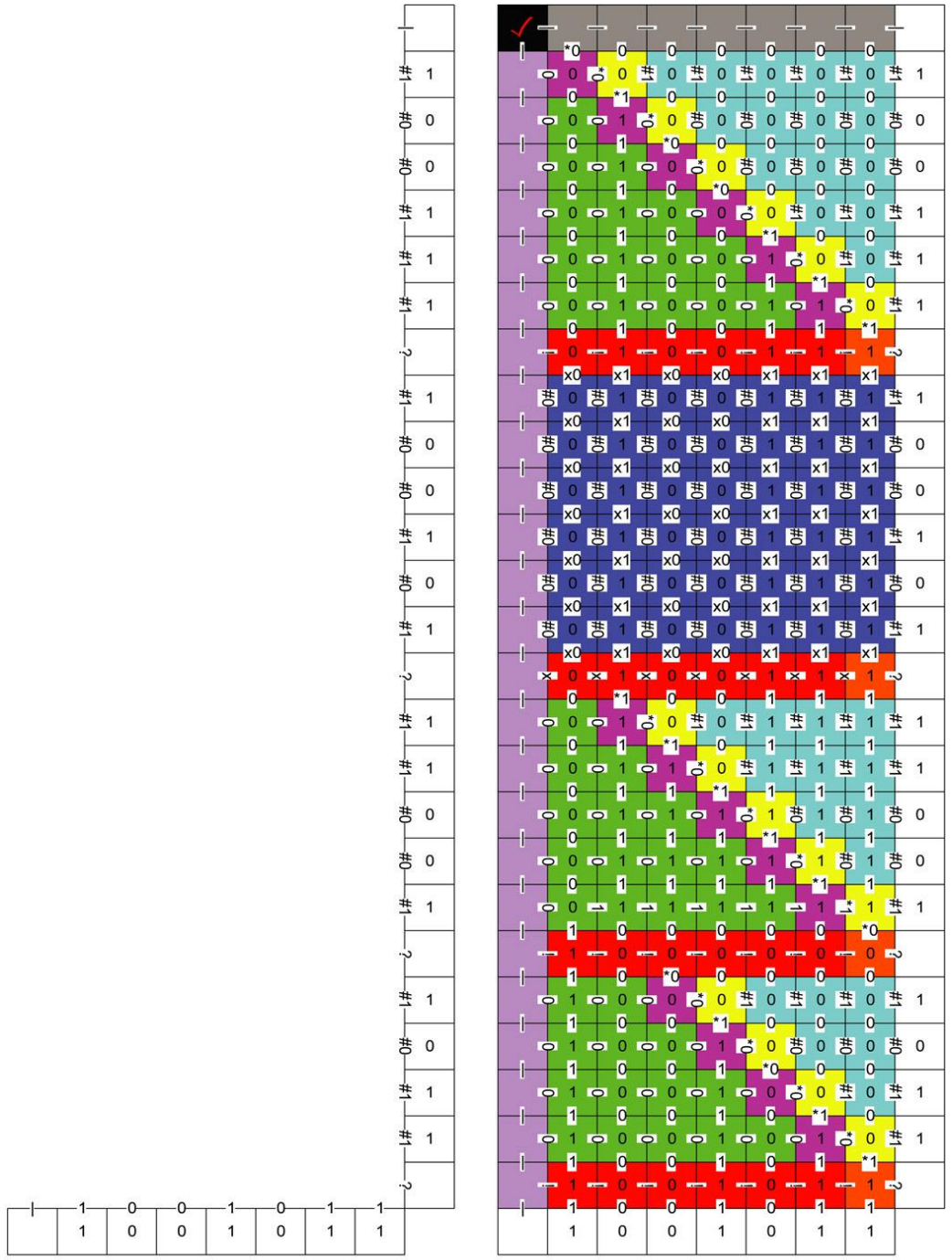
- Thus, one of the two will be **randomly chosen during the self-assembly process** leading to a **non-deterministic computation**
(which is the whole point of this experiment)

Final decision tiles Γ_{SS} and $T_{\checkmark}$

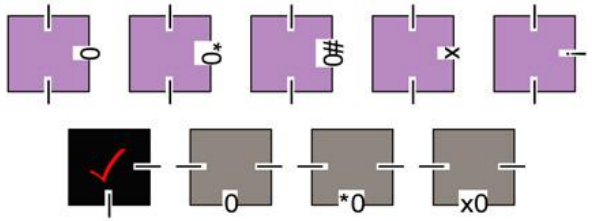


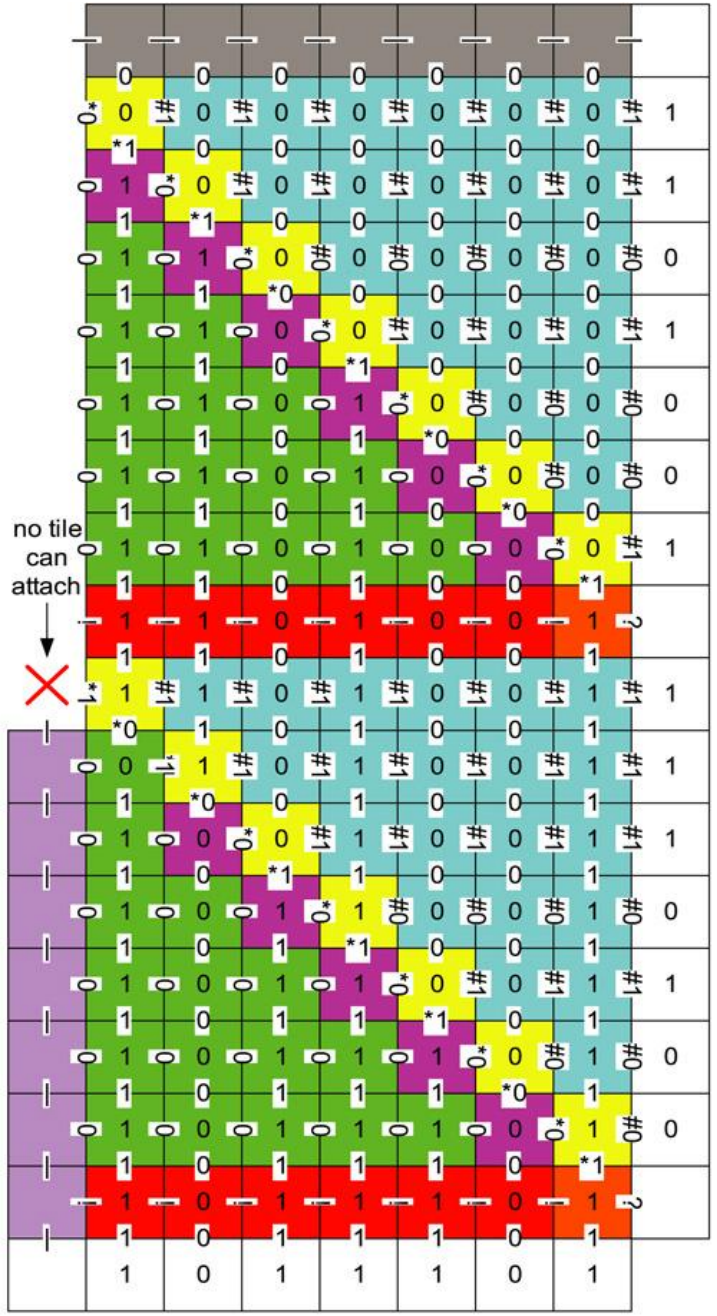
- These tiles are attached at the end of the computation
- Γ_{SS} is used to define the seed, including the tile  which triggers the non-determistic choice
- **The black tile gets attached if and only if the Subset Sum is satisfied**
- If the calculation leads to a negative value then **no more tiles gets attached** and the algorithm halts

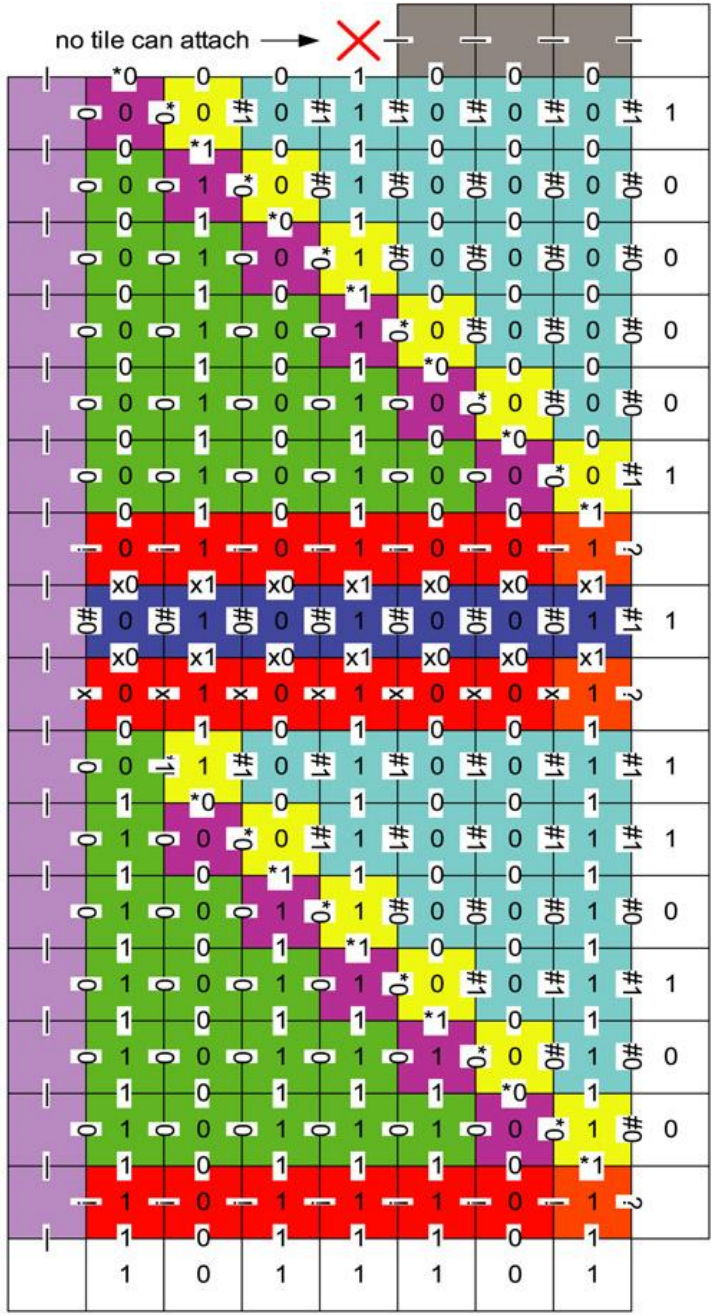
One successful execution



Unsuccesful cases







Other examples of DNA tiles application

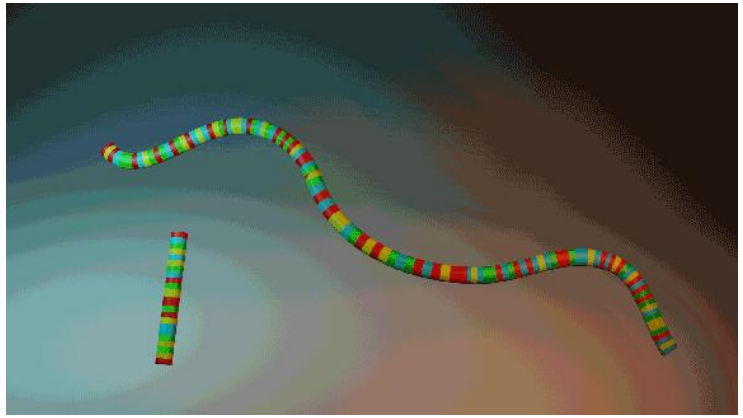
- Timetable problem [Cheng et al., Int J Comput Commun Control 2010]
- Image criptography [Patel et al., Int J Adv Res Engin Tech 2018]
- Knapsack 0-1 [Cui et al., Symposium Neural Networks 2009]
- Maximum clique [Cui et al., Advances in Comp Intell 2009]
- Graph Coloring [Zhang et la., J Comp Theor Nanosci 2009]
- Multidimensional Knapsack [Cheng et al., J Comp Theor Nanosci 2010]
- SAT [Brun, Natural Comput 2012]
- …

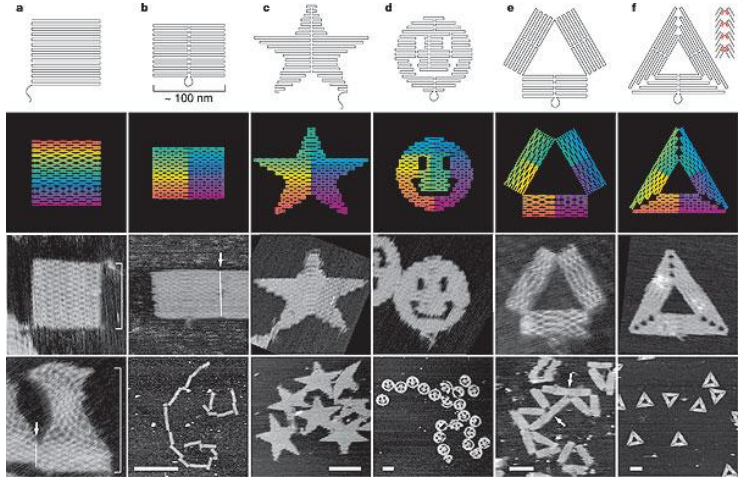
DNA origami

- Nanoscale **folding of DNA** to create arbitrary 2D and 3D shapes

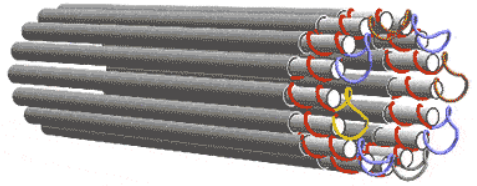
- **A single strand** is folded in a **specific position** by exploiting Watson-Crick complementarity

- Generally, the single strand to be folded is the **genome of the M13 virus** (7249 bases long), while the **DNA «staples» are created in laboratory**

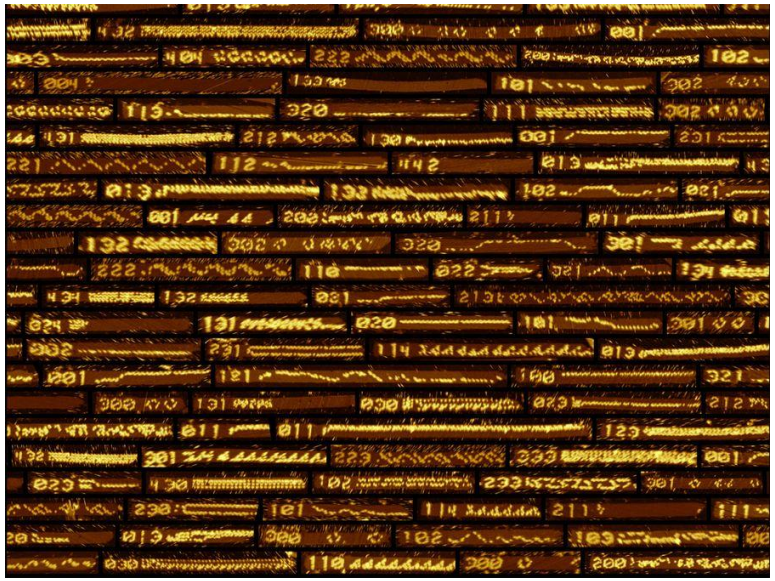




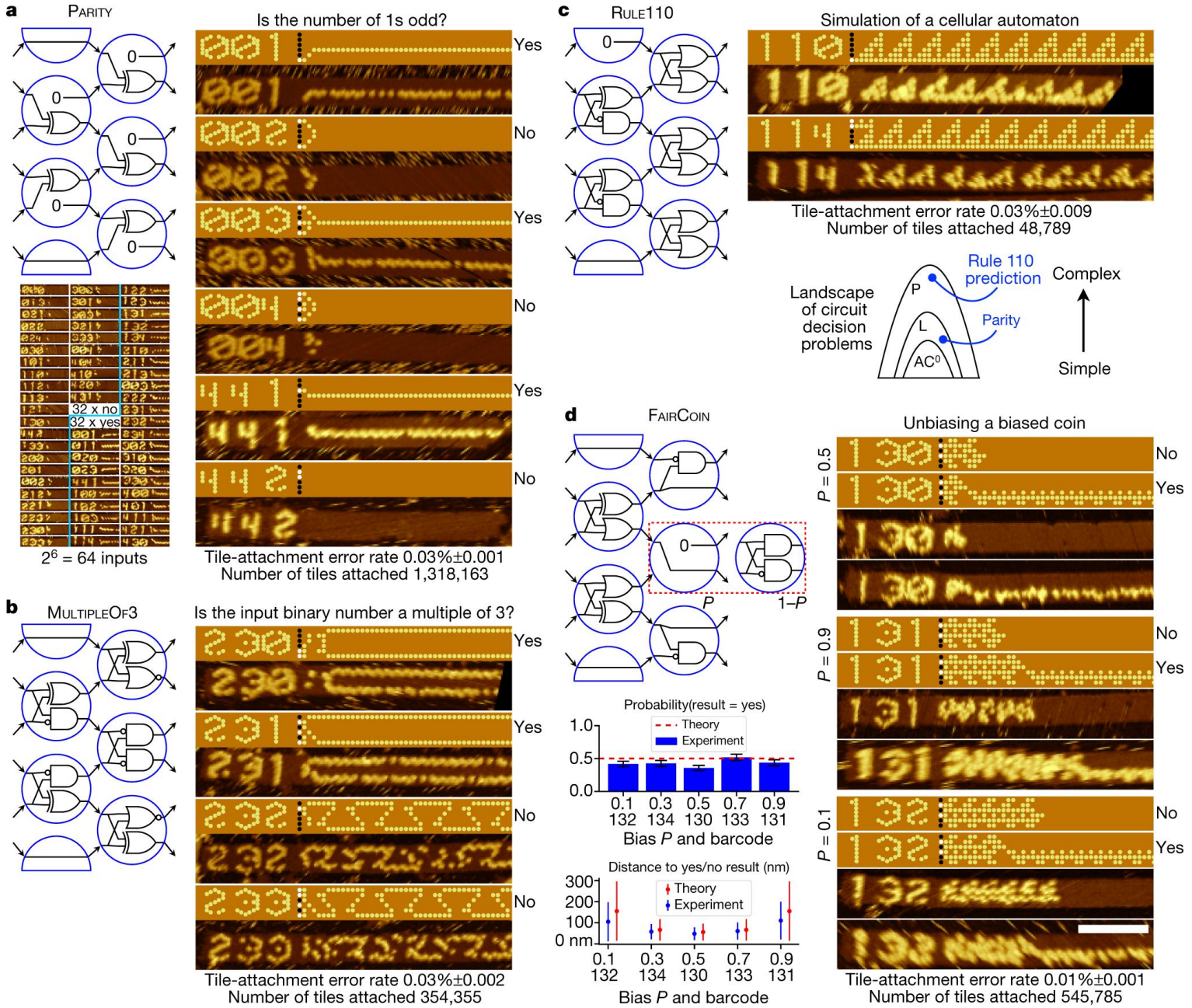
Programmable DNA computers



- [Woods et al., Nature 2019]
- **Seed** (gray): **DNA origami** attached to the the **input (red tiles)**
- Blue, brown, and yellow tiles are **logic gates**
- **1 gate == 4 tiles**
- Overall **355 types of tiles**
- **4 strands of sticky ends** to reduce error
- The algorithm is executed as the system spontaneously grows
- Source code is «compiled» to DNA
- ~1 day for computation to complete



Examples


a
Parity
Is the number of 1s odd?
Yes
No
Yes
No
Yes
No
32 x no
32 x yes
2^6 = 64 inputs
Tile-attachment error rate 0.03%±0.001
Number of tiles attached 1,318,163
b
MultipleOf3
Is the input binary number a multiple of 3?
Yes
Yes
No
No
Tile-attachment error rate 0.03%±0.002
Number of tiles attached 354,355
c
Rule110
Simulation of a cellular automaton
Tile-attachment error rate 0.03%±0.009
Number of tiles attached 48,789
Landscape of circuit decision problems
P
L
AC^0
Rule 110 prediction
Parity
Complex
Simple
d
FairCoin
P
1–P
Probability(result = yes)
Theory
Experiment
1.0
0.5
0.0
0.1 0.3 0.5 0.7 0.9
132 134 130 133 131
Bias P and barcode
Distance to yes/no result (nm)
Theory
Experiment
300
200
100
0 nm
0.1 0.3 0.5 0.7 0.9
132 134 130 133 131
Bias P and barcode
Unbiasing a biased coin
P = 0.5
No
Yes
P = 0.9
No
Yes
P = 0.1
No
Yes
Tile-attachment error rate 0.01%±0.001
Number of tiles attached 545,785

Payload releasing nanobots

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Self-assembly of a nanoscale DNA box with a controllable lid

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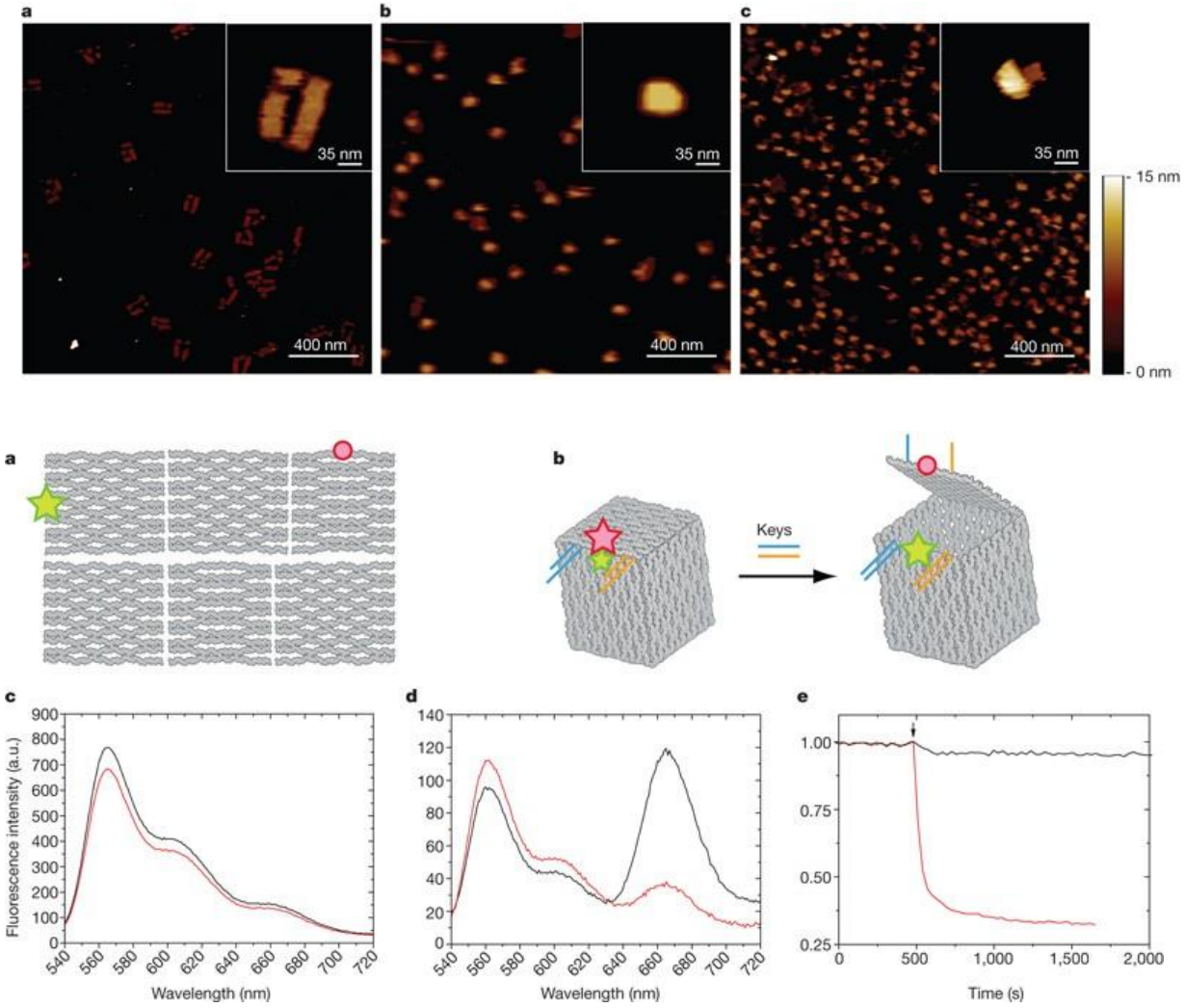
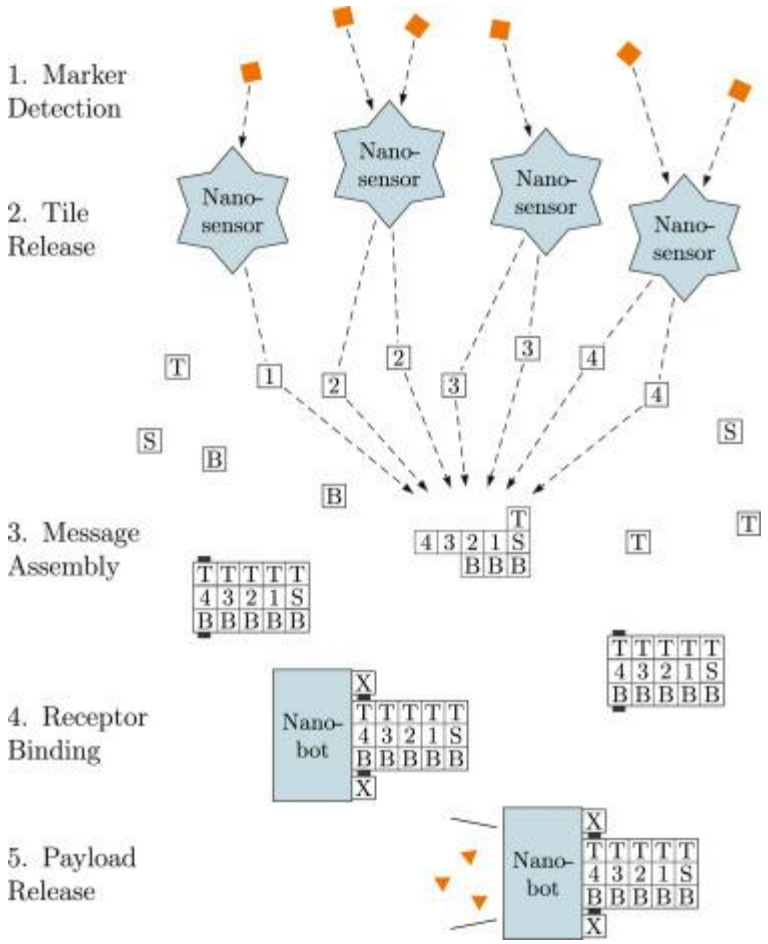


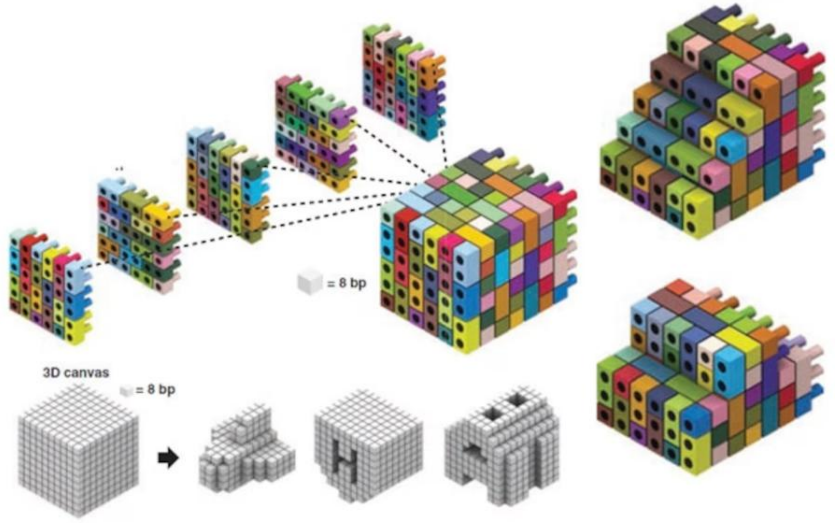
Performing computation in living organisms

- [Lau et al., Nano Communication Networks 2019]
- Nanosensors «feel» the presence of biomarkers and **release WT**
- **The WT assemble** to form a **message**
- The message is received by a more **complex nanobot** that releases a paylod (say, a drug)



From tiles to bricks

- We can also build **«LEGO bricks» with DNA** which **spontaneously assembly**
- We can **program the sticky ends** to produce virtually any **3D shape**
- [Ke et al., Science 2012]



Conclusion

- DNA is a powerful molecule: strong, regular, programmable, excellent for both storage and computation
- It provides a means to create (programmable) nano-artifacts, which spontaneously generate through self-assembly
- We can create nano-computers, operating even in living organisms able to solve NP-complete problems thanks to non-determinism and massive parallelism